United Nations Development Programme

PROJECT DOCUMENT

Nepal

Date: Project Title: **Renewable Energy for Rural Livelihood (RERL)**
Project Number: 00117173
Implementing Partner: Alternative Energy Promotion Centre (AEPC)
Start Date: 1 April 2020 **End Date:** 31 December 2024 **PAC Meeting date:** 27 September 2022

Brief Description
In recent decades, Nepal has made remarkable progress in providing electricity access to its people. Today, about 95% of the population has access to electricity. However, most of the households have only Tier 1 or Tier 2 level access and about 68% of the households still depend on traditional biomass energy for cooking. Electricity is mainly used for lighting and operating low power appliances. While progress has been made in dissemination of off-grid renewable energy systems, affordability of up-front costs (due both to high costs and lack of capital), financial sustainability (due partly to low utilization), and lack of technical capacity constrain progress. On the other hand, Nepal in spite of huge hydropower and solar energy potential, is importing over half of its electricity requirement. In this context, the Renewable Energy for Rural Livelihood (RERL) project will focus on both community-scale off-grid renewable energy technologies to promote income generating opportunities in rural areas, and larger renewable energy systems for grid connection to contribute to distributed generation. The project's core strategy will consist of five interrelated concepts: (1) provision of technical assistance for both on-grid and off-grid renewable energy projects, (2) support for increased contribution of renewable energy in the national energy mix, (3) support for improved access to clean cooking and, (4) policy inputs from lessons learned and good practices to create a conducive environment for low-carbon development path. In total, the project aims to reach more than 50,000 households, providing them with increased access to renewable energy and, as a result, improved access to economic opportunities while also contributing to the increased contribution of renewable energy in the national energy mix of the country. The project has a strong focus on rural communities and vulnerable groups and will pay particular attention to participation of women and excluded groups across all activities. The RERL project will be implemented by the Alternative Energy Promotion Centre (AEPC) under the Ministry of Energy, Water Resources and Irrigation (MoEWRI) following the National Implementation Modality (NIM) Guidelines, as agreed between the Government of Nepal (GoN) and UNDP.

Contributing Outcome (UNSDCF, RPD or CPD):
Country Programme Outcome 1: *By 2027, more people, especially women, youth and the most marginalized and poor, increasingly benefit from and contribute to building an inclusive, sustainable, climate-resilient and green society and reduced impacts of disasters at federal, provincial, and local levels.*
Country Programme Output 3.4: *Reliable, affordable, and efficient clean energy solutions promoted*
Indicative Output(s) with Gender Marker:
Output 1: Gen 2
Output 2: Gen 2
Output 3: Gen 2

Total resources required:		**US$ 29,760,000**
Total resources allocated:		**US$ 19,980,000**
	UNDP TRAC:	*US$ 1,160,000*
	ADB (TA):	*US$ 1,500,000*
	Govt. of Japan:	*US$ 2,320,000*
	ADB:	*US$ 15,000,000*
	In-Kind:	-
Unfunded:		***US$ 9,780,000***
	Government:	*US$ 5,080,000*
	UNDP:	*US$ 2,500,000*
	Others:	*US$ 2,200,000*

Agreed by (signatures):

Government	UNDP	Implementing Partner
Ministry of Energy, Water Resources and Irrigation (MoEWRI)	United Nations Development Programme (UNDP)	Alternative Energy Promotion Centre (AEPC)
Sanoj Kumar Dev Joint Secretary	**Bernardo Cocco** Resident Representative a.i.	**Dr. Madhusudhan Adhikari** Executive Director
Date: 1 February 2023	Date: 1 February 2023	Date: 1 February 2023

I. DEVELOPMENT CHALLENGE

In recent decades, Nepal has made remarkable progress in providing electricity access to its people. Despite huge hydropower and solar energy potential, Nepal is still importing over half of its electricity requirement. Latest figures published by the GoN show more than 95% of the population has access to electricity. However, most of the households, especially in rural areas, have only Tier 1 or Tier 2 level access. Electricity, for majority of these population, is mainly used for lighting and operating low power applications. The utilization of electricity in off-grid areas are still hovering around a meagre 15-20% with many individual plants still utilizing as low as 5% due to lack of enough productive energy uses (PEU) and use of more efficient appliances available in the local market. On the other hand, some isolated off-grid micro hydro plants have load factors as high as 70% throughout the day and resorting to load shedding during peak hours primarily due to large numbers of productive energy use applications and widening distribution network to nearby areas and villages.

While significant progress has been made over the last few decades in expansion of the national electricity grid and dissemination of off-grid renewable energy (RE) systems, although few, there are pocket areas in rural Nepal that are neither covered by the national electricity grid nor connected to existing off-grid RE systems. To ensure 100% connectivity to electrical services, there are ample opportunities for development of off-grid RE systems to cater the energy needs of the last mile populace within the country. However, affordability of up-front costs (due both to high costs and lack of capital), financial sustainability (due partly to low utilization), and lack of technical capacity constrain progress.

Further, the unreliable and inadequate nature of the national electricity grid means there is a huge challenge in fulfilling the cross-sectoral energy needs of the country. With more than 70% of the agricultural land in Nepal still unirrigated, there is also an immense opportunity for pumping water for agriculture through isolated RE systems. Farmers have traditionally been preferring fossil fuel-based pumping systems for water pumping mainly due to the lack of knowledge on other options and, in many cases, high upfront cost of RE-based pumping systems. According to a recent study, there is an opportunity to reduce the emissions of greenhouse gases through fuel switching. AEPC has successfully promoted around 1500 solar pumping systems, but such efforts need further consolidation and a more streamlined approach for wider benefit especially for small farm holders with low purchasing power. Similar challenges arising due to lack of, or unreliable supply of electricity do exist in health and education sectors. In the context of COVID-19 pandemic, the need to ensure supply and reliability of electricity to health centers throughout the country is even more crucial.

Recent national data also show that about 68% of the households still depend on traditional biomass energy for cooking, a task which women is traditionally responsible for. Although Nepal is on course of meeting Sustainable Development Goal (SDG) 7 and SE for All commitments on electricity access for lighting, the country is facing a huge challenge in providing cleaner cooking solutions to its people. Electric cooking in both urban and rural areas in Nepal is an especially big challenge as there are limits to power access and the cost of electricity is relatively expensive compared to cooking gas.

Hence, there are still many areas within the RE subsector of the country that needs further interventions especially in, (i) installation of new RE power generation projects in mainly rural areas to cater to the electricity needs of pocket areas not covered by either the national grid or existing off-grid projects, (ii) promotion of PEUs and provision of electricity-based livelihood options, (iii) integration of off-grid RE systems to the national grid to reap expected benefits of distributed generation and increasing the share of RE in the national energy mix of the country for energy security and sustainability of past efforts on improving electricity access to the rural population, (iv) improved energy reliability and modernization of cross-sectoral applications such as solar pumping, energy for health and education, etc., (v) contributing to the global fight against COVID-19 pandemic, (vi) capacity development for RE development in the country targeting all stakeholders including the local and provincial governments who are responsible for RE promotion in the country after the promulgation of the new Constitution of Nepal in 2015.

In this context, the Renewable Energy for Rural Livelihoods (RERL) will focus both on community-scale, off-grid RE technologies to promote income generating opportunities in rural areas, and larger RE systems for grid connection contributing to distributed generation as well as reduction of greenhouse gas emissions. The project will support to find solutions for cooking utilizing different types of technologies.

The Government of Nepal (GoN) through various policy documents, including the 2018 Ministry of Energy, Water Resources and Irrigation (MoEWRI) White Paper 'Current Status and Future Course of Energy, Hydropower and Irrigation Sector' and the GoN's 15th Plan (2019/20 – 2023/24) has emphasized promotion of RE both for providing

electricity access to off-grid areas for economic development and to reduce dependency on imported fossil fuels. The project will directly contribute to GoN's commitments, as well as SDG 7: Affordable and Clean Energy, and SDG 13: Climate Action, by ensuring access to affordable, reliable sustainable and modern energy for all through promotion of RE for remote off-grid areas not being currently served by the national electricity grid, and to feed the national grid. Likewise, the project will contribute to substantially increasing the share of RE in the national energy mix. With promotion of RE as the entry point, the project will also contribute to SDGs 1: No Poverty, 5: Gender Equality, 8 Decent Work and Economic Growth, and 12: Sustainable Consumption and Production, as well as several other SDG targets.

II. STRATEGY

UNDP has been supporting the Government of Nepal for the past two and half decades for the promotion of RE in the country through the implementation of Rural Energy Development Programme (REDP). As mentioned previously, there have been significant progress in the access of electricity figures from the time UNDP-supported REDP to this time. While electricity generation and connecting households to electricity was the priority two and half decades back, the need to sustain and upscale past achievements and impact the livelihoods of people through the generated electricity is the current necessity. The latest UNDP-supported project, RERL, focused on these and were able to make significant progresses on these aspects. However, there are still plenty to achieve especially relevant considering the geo-political changes that took place in Nepal with the promulgation of the Constitution of Nepal 2015. With the local and provincial governments given exclusive rights for RE development in the country, the role of external support in this sector has also evolved while creating new opportunities as well as challenges further strengthening whatever has been done and achieved in the past.

The project will work closely with several key stakeholders, including AEPC and other federal level government agencies, NGOs, private sector actors, and provincial and local governments to enhance their capacities to be able to plan, implement, operate and manage RE systems and PEUs. The specific strategic focus of the project are illustrated below;

a) Working with the local governments

The Constitution of Nepal 2015 and the Local Government Operation Act 2017 have given mandate for development of alternative/renewable energy and hydropower up to 1MW to the local governments. Consequently, it is imperative to maintain close cooperation and collaboration with the local governments for development of mini hydro and solar PV projects. UNDP has over 20 years' experience of working closely with the local governments for RE promotion through its flagship projects, REDP and RERL. For example, RERL have prepared methodologies and helped seven municipalities to prepare Municipal Energy Plans (MEPs). In the next phase, the partner municipalities will also be supported to prepare their energy plans that will help identify energy gaps, resources and budget required to provide clean energy solutions to all as envisaged by SDG 7. The inclusion of pipeline projects into the MEPs will ensure value addition and further strengthening of South Asia Sub-Regional Economic Cooperation (SASEC) activities at the field level. Municipalities will be encouraged to provide equity investment in projects. This would create a favorable political climate for project implementation thus reducing risks to the projects and ensure increased stake of local governments ultimately improving sustainability of these projects. RERL will look to support projects up to the stage when the local governments are capable of managing the RE projects. For this, active participation of the local governments will be ensured and also extensive package of capacity development for local governments will be undertaken to reach a level where the local governments are strengthened to be able to manage the RE projects under its jurisdiction.

b) Improving policy environment

UNDP, through its programs including RERL, has been assisting the GoN in formulating policies, guidelines and manuals for institutionalizing RE best practices. The past learnings from RE development have been reflected in the Rural Energy Policy 2006, RE Subsidy Policy 2016, and RE Subsidy Delivery Mechanism 2017. While the primary objective of the proposed technical assistance (TA) is to ensure implementation of field level activities for completion of projects, RERL will also ensure that lessons learnt from field level implementation are internalized and institutionalized through policy frameworks at the federal, provincial and local level. The learnings and the process followed will be documented and recommended to AEPC and MoEWRI as policy inputs for wider replication.

c) Institutional development and strengthening

One of the key barriers for attracting financial investments in off-grid RE projects is the lack of credible institutions at the subproject level for sustainable operation and management. Banks and other potential investors are averse to invest in community-based projects due to their perception of increased risks without accountable institutions at the individual project level. Financial institutions perceive the notion of "everybody's responsibility is nobody's responsibility" as a key risk in community owned projects. RERL will support the beneficiary communities to establish either cooperatives or companies as Special Purpose Vehicles (SPVs) with accountable governance and transparent and incentive-based management for project development and operation.

d) Gender Equality and Social Inclusion (GESI) mainstreaming and Environmental and Social Safeguard (ESS)

RERL accords high priority towards mainstreaming GESI aspects into every step of the project cycle. The project will not only ensure participation of women and members of disadvantaged groups during the implementation of project activities but will also ensure that they maximize benefits from access to electricity. As required by the government, 33% participation of women in decision making bodies and 33% participants in capacity development activities as required by the project will be ensured. RERL will prepare and ensure that the GESI Action Plan is implemented at all stages of the project cycle. Similarly, it will prioritize gender responsive and socially inclusive community consultations and engagement to better understand concerns of different groups in the communities in the project areas while also enhancing knowledge of the beneficiaries on GESI issues. In particular, RERL will prioritize the engagement of women and members of vulnerable and disadvantaged groups in the PEU activities of the project.

RERL accords high priority for drudgery reduction of women as part of mainstreaming gender equality and social inclusion into the project. As women are traditionally responsible for household cooking in the Nepali context, the project believes interventions in domestic cooking can actually positively impact the lives and well-being of women and can definitely contribute towards engaging women to more non-traditional gender roles. The project intends to provide cleaner and efficient cooking solutions mostly in off-grid areas shifting away from the traditional firewood-based cooking to alternatives, more specifically electricity-based cooking so as to reduce not only the time taken for cooking and but also reduce the health implications of using inefficient and dirty fuels for cooking. The promotion of electricity-based cooking not only improves the lives of women on the demand side but also greatly contributes on the supply side by improving the utilization of power and revenue thereby aiding sustainability as discussed earlier.

Similarly, RERL will ensure management, oversight and monitoring of implementation of Environmental and Social Safeguards (ESS) during the preparation and implementation of the RE projects. RERL, particularly in relation to implementation of ADB-funded SASEC project, will help implement different ADB frameworks including Environmental Assessment Review Framework (EARF), Indigenous People's Planning Framework (IPPF) and Resettlement Framework (RF) of the project while also developing capacity of project staff and stakeholders on the application of these Frameworks. As an important facet of project development, RERL will review all Detailed Feasibility Studies (DFS), Detailed Engineering Designs (DED), Environmental Management Plans (EMP), and Initial Environmental Explorations (IEE) with due diligence to verify that environmental mitigations are duly addressed during project planning and implementation. RERL will ensure that developers consult the affected people and stakeholders during implementation of RE projects. It will also ensure that the environmental safeguard compliance monitoring is conducted as per ADB Safeguard Policy Statement and national regulations and requirements.

e) Productive use of energy and sustainability

Past experiences have demonstrated that rural communities require technical support not only during RE project formulation and implementation but even more so after completion of installation. Post installation support will be an integral part of RERL TA. Post installation support will include both institutional strengthening for smooth operation of the plants and promotion of PEU. SPVs will be encouraged and supported to establish PEU with connected load of at least 20% of the installed capacity of RE electrification projects. To achieve the target of PEU, RERL will work closely with AEPC to provide TA to projects to identify potential enterprises and entrepreneurs, business plan preparation, skill development, market linkages and financing, including government subsidies and credits.

f) Private sector involvement

The Government has made provisions for involving the private sector in development of renewable energy projects in the country with the expectation that the private partners will bring in efficient management to energy projects and help achieve sustainability. However, the involvement of the private sector in the smooth and sustainable operation,

of RE projects is currently insignificant. The role of the private sector is limited to installation of systems. On one hand, the private sector lacks enough incentives to venture into post installation service provision in the rural areas while it also seemingly lacks adequate technical expertise needed for smooth and sustainable operation and maintenance of renewable energy systems. The project will pursue a strategy to develop capacity of the private sector including trained human resources and will also encourage for increasing private investment into the renewable energy sector in the country.

g) Climate Resilience

Nepal is considered as one of the most vulnerable countries in the world to climate change induced disasters primarily flood, landslide and drought. The Disaster Risk Reduction and Management (DRRM) thematic areas of the National Climate Change Policy 2019 emphasizes reducing the impacts of climate induced disasters on sources of clean energy mainly hydropower with the involvement of local and provincial governments. Both existing MHPs and new RE systems will be refurbished/developed to withstand climate hazards and contribute towards both climate change mitigation adaptation.

III. RESULTS AND PARTNERSHIPS

Expected Results

The main objective of RERL is to improve livelihoods of rural communities/vulnerable groups through increased access to renewable energy sources for productive use thereby increasing the share of renewable energy in Nepal's energy mix for a low-carbon development path to support, through collaboration with AEPC, Provincial and Local Governments, the private sector and the civil society. To achieve this objective, the project will provide technical assistance to achieve the following outputs;

Output 1 – Improved access to clean, affordable and reliable energy solutions

Under this output, technical support will be provided to AEPC, Provincial and Local Governments, rural communities and the private sector to develop larger RE projects for improved access to clean, affordable and reliable energy solutions to energy-deprived communities of the country.

Activity 1. Technical assistance for promotion of mini hydro and solar mini grid systems: RERL will support AEPC to develop 4.3 MW of mini hydropower projects and 0.5 MW of solar mini grids under the SASEC project (funded project), which is financed by the Asian Development Bank (ADB).

RERL, since its inception in 2014, has been providing technical assistance to the AEPC including the SASEC project. After the phase-out of external funding in National Rural and Renewable Energy Programme (NRREP) in 2017, RERL has been providing significant technical assistance to the SASEC project under the complementarity approach to meet the objectives of both the projects. As of March 2022, RERL has provided technical support to the SASEC project to successfully install a total of nine solar mini grids with a cumulative generating capacity of 565 kWp, against the project target of 500 kWp, benefitting 1,612 rural households previously devoid of electricity access. Similarly, RERL has supported to successfully commission 4 mini hydro projects with a total generating capacity of 1,100 kW that has provided electricity access to 3,009 rural households till date with more being connected over time. Since the SASEC project target of mini hydropower projects is 4.3 MW, there is still a lot to be delivered to meet the project targets.

In this context, RERL will provide support at all stages of the project cycle before, during and post construction stages, focusing on identification of best available technologies, quality, standards and sustainability including promotion of productive energy uses and livelihood options. RERL support at different stages will include the following:

Pre-construction activities

- Identification of RE projects based on best available technology
- Feasibility study, including GESI and environmental assessments, and detailed engineering design
- Community mobilization, institution establishment and strengthening including promotion of micro finance activities
- Financial closure in close coordination with Central Renewable Energy Fund (CREF)
- Procurement of equipment and contractors

During-construction activities

- Construction supervision of RE projects ensuring quality and compliance of standards
- Liaison between project developers and contractors for timely completion of the construction works
- Testing and commissioning of projects
- Support implementation of Environment and Social Safeguard (ESS) management activities

Post-construction activities

- Technical trainings for operators, linemen, house wiring, etc.
- Efficient and transparent operation and management of projects by providing training, exposure visits, 'on-the-job' training, etc. on office management, administration, computerized financial management system, human resource management, inventory management, etc.
- Establishment and operation of sub-committees for Local Economic Development, Community Energy Services, Saving and Credit Management and Grievances Handling and Conflict Management
- On-the-job training for operation and management
- Tariff collection and management
- Peer-to-Peer (P2P) learning wherever possible
- Knowledge exchange, experience sharing and technology transfer
- Documentation for Power Purchase Agreement with Nepal Electricity Authority (NEA), if need be.

Similarly, RERL will also support activities related to ensuring community participation in off-grid RE projects providing clean energy access to rural communities and vulnerable groups. The comprehensive social mobilization practiced by RERL instills transparency, accountability and participatory decision making. Another important feature is active participation of community members, including both men and women, and persons from excluded groups, during the post installation stage, to ensure local ownership and sustainability. In the context of RE projects in urban areas and for grid interconnection, private sector participation and investment will be encouraged and supported to establish Special Purpose Vehicles (SPVs) and/or Energy Service Companies (ESCOs). To achieve this, the following sub activities will be undertaken:

Social Mobilization

- Mobilization of communities, including both men and women, as well as persons from different excluded groups, following the comprehensive Community Mobilization Package of REDP/RERL
- Facilitate to raise equity and engage communities at all stages of the project cycle
- Raise awareness, disseminate information and create demand for PEU
- Facilitate for active participation of women and excluded groups in project management and to engage in micro finance activities in order to maximize benefits from access to electricity

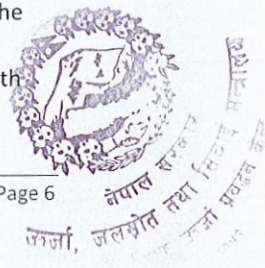
Institution Establishment and Strengthening

- Orientation on different institutional arrangements for project development and operation such as cooperative, company, lease out model, etc.
- Facilitate establishment of institutions fulfilling all legal requirements
- Facilitate to form sub/committees with clear roles/responsibilities for efficient management
- Develop and implement project operation and management guidelines for smooth operation and management
- Introduce computerized accounting and financial management systems to increase transparency and accountability

Coordination with Local Governments

- Facilitate for local government's equity investment in RE projects
- Support to establish linkages between RE projects and respective local governments
- Facilitate towards resolution/management of any community issues in the RE projects in close coordination with the respective local governments

To support AEPC to implement the SASEC project, RERL will support social mobilization while also ensuring that the communities are capable and prepared to operate and manage the RE projects through establishment and strengthening of appropriate institutional structures. It is envisaged that RERL will maintain close engagement with



project developers and stakeholders for at least six months after completion of physical installation of RE projects through dedicated field facilitators in each RE project, deputed based on needs as far as possible. RERL will provide support to establish and strengthen 18 legally registered institutions to develop and operate nine mini hydropower projects and nine solar mini grids.

Furthermore, the project will aim at supporting more productive and efficient utilization of the RE generated from the projects. Promotion of PEU will greatly help to improve utilization factor of these projects, as well as ensure operational sustainability through improved revenue generation. The cooperatives and ESCOs/SPVs will be supported for identifying potential uses of electricity and potential for entrepreneurship, prepare business plans, establish PEU enterprises, skill development and market linkages. The main sub activities are:

Mechanism Development

- Establishment of Local Economic Development Committee for promotion of PEU
- Engagement of Savings and Credit Groups for promotion of PEU
- Establishment of mechanism for tariff fixation and collection for enterprises
- Establishment of mechanism to access government subsidy for PEU

Business promotion and Entrepreneurship Development

- Business Opportunity Assessment to identify economic and business opportunities
- Entrepreneurship development training
- Support to the preparation of business plans
- Facilitation for accessing financial services including government subsidies
- Facilitate for product diversification/quality assurance and market linkages
- Support entrepreneurs to establish linkages with regional/local business development service providers as well as financial institutions

Innovations

- Explore potential new uses of reliable electricity supply and provide engineering support for fuel switching from traditional fuels to RE
- Support for promotion of electric cooking, space heating/cooling, water heating, etc.
- Support to promote energy efficient appliances

RERL will support AEPC/SASEC to establish productive end uses with total installed capacity of 980kW (20% of total project capacity).

Activity 2. Increase utilization of energy from micro hydro systems in remote rural off-grid/on-grid areas: In Karnali and Sudur Paschim provinces, due to very narrow network of grid electricity given the remoteness and difficult mountainous terrain, access to electricity to majority of the population is limited. Even the potentials of existing off-grid micro-hydro power plants for productive end-use are not optimally utilized as most of the plants run below their production capacities at plant factor of 15% on average. These two provinces are also far behind in terms of socio-economic indicators and are hard hit by current pandemic. The opportunities lie in these two provinces to operate the micro-hydro plants to their full capacities; utilize potentials of productive end-use by increasing the plant factor; and improve quality and reliability of electricity for sustainable enterprise development through:

- Support to create livelihood opportunities for the poor and marginalized people through supply of reliable electricity and enterprise development by increase in plant factor up to 50% and improvement in quality of electricity.
- Provisions of lift irrigation, agro-processing, micro-watershed management and establishment of natural resource-based enterprises to contribute to livelihood and building resilience.
- Support for application of energy-based technologies for value chain development and e-cooking up to 15% of households through technical feasibility and demand side management to create local employment
- Launch of massive awareness programme to replace use of (a) biomass for cooking by e-cooking, and (b) fossil fuel by electric-based alternatives for lighting, cooking, and agro-processing.
- Nepal has a long experience of developing micro hydropower projects of less than 100kW capacity and there are over 2000 such systems spread throughout the middle hills and mountainous region of the country. There is capacity to design and build MHPs within the country. However, electricity in rural areas is mainly used for lighting for a few hours in the evening and the plants do not generate enough revenue for

sustainable operation. It is proposed here to help rural communities to upgrade/refurbish their existing MHPs, make them climate resilient, promote productive uses for income generation, utilization of electricity for cooking and space heating and grid interconnection. It is expected that with these supports the plants will generate enough revenue to not only cover their expenses but surplus for future repair and major maintenance if required. The plants will then be able to provide reliable and quality electricity at affordable rates and utilization of electricity for improvement of livelihoods including reducing drudgery of women and children will be realized. Furthermore, these supports will ensure that the national resources provided for construction of these projects will be utilized for extended period.

Other sectors such as health and education within Nepal suffer largely through either lack of energy or, in most cases, challenges in reliability of electricity supply. There are plenty of opportunities in improving system reliability and modernization of cross-sectoral applications of RE. Under this output, RERL will focus on wider cross-sectoral applications of RE to ultimately improve livelihood and well-being of the local communities.

Promotion of distributed generation: The idea of distributed generation (DG) is installing power generation systems near the load centers usually grid interconnected small-scale renewable energy plants to improve the voltage of the distribution system, enhance reliability of the system and reduce line losses. By injecting power locally, it also helps to minimize the infrastructure cost of long transmission and distribution systems. Further, by generating revenue harnessing indigenous resources, DG contributes to the local economy and more equitable growth. Realizing these benefits, many utilities around world are moving towards distributed generation from their centralized systems with large power plants.

In case of Nepal, realizing distributed generation is made easier by the fact that there are over 400 isolated micro hydropower plants (MHP) of more than 25 kW capacity and over 135 MHPs of more than 50kW. with total capacity of 11MW that could be interconnected with the national electricity grid. These MHPs have their own distribution networks and provide electricity access to about 200,000 households which could be easily brought under the Integrated Nepal Power System (INPS). On the other hand, the Nepal Electricity Authority (NEA), the grid owner, has very long distribution lines that face frequent outages and have very low voltage levels, could be strengthened by injecting generation near the load centers.

Furthermore, distributed generation could be designed to help strengthen federalization by involving local governments, communities and entrepreneurs as envisaged by the Constitution of Nepal 2015 and mandated by the Local Government Operation Act of 2018 that gives overall responsibility to the local governments for development of hydropower plants up to 1 MW and rural electrification. Developing small-scale generating capacity at the local level will not only create jobs locally but also could be a reliable source of revenue for local governments. AEPC, from the past, have received significant demands from municipalities primarily during the preparation of Municipal Energy Plans for support to develop grid connected mini hydro and solar PV projects both for local electrification needs and additional revenue generation for the municipality. As AEPC has limited capacity in grid connected systems, RERL will support the municipalities and AEPC with technical assistance to design, implement and commission such larger grid connected systems. RERL has assisted AEPC in the past to interconnect smaller RE systems into the grid both in preparing documentations for grid interconnection and also designing of such systems. Building on the past experiences, RERL will further assist the municipalities and AEPC to develop modalities and also coordinate closely with NEA for development and promotion of grid connected larger renewable energy projects. As a major starting point of development of larger grid connected systems under the concept of distributed generation, RERL will focus to revive and grid connect mini and small hydropower systems in district headquarters which are owned by NEA.

The major sub activities that will be carried out under this are:

- Identification of RE projects based on best available technology for grid interconnection
- Feasibility study and other technical studies, including GESI and environmental assessments, and detailed engineering design
- Engagement of local stakeholders including the local governments, NEA, beneficiaries, private sector and banks and financial institutions into project development and operation
- Documentation for Power Purchase Agreement with NEA and NERC
- Financial closure in close coordination with Central Renewable Energy Fund (CREF)
- Technical assistance during the procurement of equipment and contractors including bid documentation
- Coordinate for timely testing and commissioning of projects
- On-the-job training for operation and management

Activity 3. Improve access to universal modern energy services for vulnerable groups through solar energy solutions: In Madhesh Province, which covers mostly the flat lands of the central terai region of Nepal, the focus will be to promote access to energy services for those who are living under extreme deprivation in the poverty-pocket areas without any alternate livelihood options available to them. Despite nearly 95% of population in Nepal has access to grid electricity, there are few poverty-pocket areas in the province which are yet not connected to national grid, and even if connected to grid, due to extreme poverty, majority of population living in those areas cannot afford the grid electricity. Current practice within the community is to burn agricultural residue, cow dung and fuelwood as prime source of energy for cooking and chose to migrate outside for labor work to support livelihood. Lack of reliable and affordable source of energy is considered to be the main barrier in creating economic opportunities for which the project plans to support through:

- Provisions of solar mini-grid and solar water pumping leading to set up green enterprises, increase in agriculture production and local employment through improved irrigation facilities, agro-processing, commercial farming and multi-layered community wood lots.
- Utilization of inherent skills of the community to sustainably operate and maintain the solar energy systems and help them draw their living by setting up micro, small and medium enterprises.
- Making use of solar power for various social and environment benefits such as drinking water, sanitation, irrigation, improved quality of education and health services, waste processing and improved internet connectivity.
- Nepal has experience of promoting both Solar Mini Grids and Solar Pumping Projects and UNDP Nepal has been closely working with AEPC to develop implementation modalities, subsidy regimes and technical assistance for installation, promotion of productive uses, etc. So far, with UNDP support, AEPC has completed 9 Solar Mini Grids with the total capacity of 565kW providing electricity access to over 1600 households and establishment/fuel switching from diesel/firewood to electricity of more than 250 productive uses. Experience shows that these solar projects are sustainably operated and maintained by rural people once the upfront cost of equipment is subsidized and their technical capacity is enhanced. Based on the experience, the government is promoting another 23 SMGs with its own resources and another 10 SMGs larger than 100kW with private sector involvement. However, all these projects are in the hilly region of the country. This proposal intends to provide support to selected communities in the Terai region to install SMGs for improvement of livelihoods. Further, recently a community in the Terai region with UNDP support completed a solar powered multiple uses of water project that provides clean drinking water to 256 households and irrigation facility to local farmers. Integrating drinking water and irrigation facilities in a single water lifting project is quite new to Nepal and this experience will be scaled up in the proposed project to help people to have both drinking water and irrigation facilities. Moreover, the private sector in Nepal is capable of providing survey, design, procurement and supply, installation and after sales services for both SMG and solar pumping projects.

Under this output, RERL will focus on cross-sectoral applications of RE systems for wider socio-economic benefit maximization to the beneficiary community. These include but not limited to the following energy applications:

Powering health and educational facilities: RERL have previously supported few health centers, especially snake bite treatment centres in the Terai region, and educational institutions with technical and financial assistance to install solar backup systems for powering essential medical equipment (vaccine refrigerators, autoclave machines, etc.) in health facilities and modern teaching equipment (laptops, projectors, e-software, etc.) in educational institutions. However, the demand for such support is ever increasing. AEPC has been supporting the activities to some extent but the demand for such support is easily being outweighed the government budget allocation. Especially in the changed context after the pandemic, the demand is set to further increase essentially to overcome the risks of transmission of coronavirus. Both AEPC and RERL will continue to support the local governments in providing renewable energy backup solutions to the local health and education facilities to improve access to modern and uninterrupted health and education services at the local level. Apart from backup solutions, the project will seek to provide other energy related services to the institutional facilities which include solar hot water systems and waste disposal/incineration systems, wherever technically and financially feasible. The project will work closely with the local governments and other local stakeholders to identify, assess and implement such activities at the field level. Further, RERL will seek partnerships with other like-minded partners for additional resource mobilization and assistance in reaching out to as many beneficiaries as possible. The major sub activities within this activity are:

- Demand collection and site identification

- Feasibility study and detail engineering design
- Technical assistance in procurement of equipment and contractors
- Construction supervision of projects ensuring quality and compliance of standards

Access to drinking water and irrigation through pumping systems: The project will promote RE based pumping systems for drinking water and irrigation. The demand and potential to promote RE-based pumping systems in this topographically challenging terrain is very high. RERL and AEPC have been receiving huge demands for pumping water for drinking which have not materialized due to budgetary constraints. Similarly, the plain of the Terai also has vast potential for promoting solar based pumping systems primarily for irrigation thus replacing the currently used fossil fuel based pumping system. Although electricity is available in mostly all areas in the Terai, there is hardly any access to electricity in the fertile fields with only transmission and distribution lines passing overhead. There is also no government interventions to providing electricity access to the irrigation fields and farmers are resorting to burning diesel to pump water for irrigation. With the use of solar power to pump water in the Terai, there is a huge applicability of renewable energy for improving agricultural productivity while also cutting down on the greenhouse gas emissions. The major sub activities under this activity are:

- Demand collection and site identification based on best available technology
- Feasibility study and detail engineering design
- Resource mobilization and financial closure
- Technical assistance in procurement of equipment and contractors
- Construction supervision of projects ensuring quality and compliance of standards
- Liaison between project developers and contractors for timely completion of the construction works
- Testing and commissioning of projects

Energy for Food Security: Nepal has improved food security status over the recent years with 4.6 million people still food insecure. However the incidences of severe food insecurity in remote rural areas especially in Karnali and Sudur Paschim Provinces is very high as the local population do not grow enough food to last them for an entire year and must depend on other sources of income to meet their food requirements for more than 9 months a year. Food insecurity in Nepal has been associated with, among others, susceptibility to natural disasters such as drought, floods and landslides and poor infrastructure including energy infrastructure. The project will contribute towards food security of mainly Karnali and Sudur Paschim Provinces by providing electricity access for modernization of agriculture and opening new economic opportunities that build resilience and strengthen livelihoods. Support will be provided for water lifting using existing or new RE systems to produce high value agricultural products such as fruits and vegetables. Similarly, energy powered storage and handling (e.g. refrigeration, drying, and packaging) reduce post-harvest food loss and improve food quality, increasing availability of nutritious foods at the household level and enabling farmers to control the timing of crop sales, improving household income. The major activities under this output are:

- Demand collection and site identification
- Feasibility study and detail engineering design
- Technical assistance in procurement of equipment and service providers
- Construction supervision of projects ensuring quality and compliance of standards
- Liaison between beneficiaries and contractors for timely completion of the works

Promotion of clean cooking technologies for health and environment: Nepal has adopted SEforAll targets of 'Electricity Access to all by 2030' for both lighting and cooking. Based on current trends, Nepal is on course in meeting the designated target for lighting but meeting the cooking target seems highly unlikely in the present condition. With projection of increased electricity generation over the next few years, NEA has been trying to promote electricity-based cooking in its catchment areas mostly in urban and peri-urban areas. However, such initiatives have not been very successful primarily due to the perceived high initial cost of fuel switching and also unreliable grid supply. Imported LPG is still the preferred option as the primary cooking fuel in urban and peri-urban areas. On the other hand, readily available firewood is the primary cooking fuel in majority of the vast rural areas of the country where many of AEPC-supported renewable energy technologies such as micro hydro, solar PV, biogas and improved cook stoves are installed for meeting lighting and cooking needs. To fulfil cooking needs, AEPC has promoted more than 400,000 household sized biogas plants and around 250 large biogas plants till the end of FY 2018/19. However, with decreasing trend in animal holding in rural areas, many biogas plants are facing issues with feedstock thereby resorting to abandoning biogas as cooking fuel. RERL, in the last phase, conducted feasibility study and piloted

electricity-based cooking in off-grid micro hydropower plants for the dual purpose of reducing the burden on dwindling forest resources and increasing the utilization of the power plant thereby improving the revenue of the plant and in turn, aiding sustainability of the off-grid plant. The project will prioritize the promotion of electricity-based cooking in off-grid areas and community-based biogas in potential areas while also promoting ICS for efficient and sustainable cooking through firewood and solid biomass fuels. The major activities under this output are as follows:

- Technical studies on the feasibility of electric cooking in larger sized RE projects such as mini hydro, micro hydro and large solar PV systems
- Orientation and awareness raising activities at the community level for wider adoption of electricity-based cooking
- Support the private sector on adaptive research and customization of electric cook stoves
- Technology transfer and technology dissemination on modern electric cook stoves including low wattage cookstoves
- Promotion of community-sized biogas plants
- Development of women and vulnerable groups as ICS technology promoters

Activity 4. Technical assistance to AEPC for implementation of programmes and projects: The implementation of various programmes and projects within AEPC are being delayed and/or hampered by the lack of sufficient technical assistance. Although the government provides financial assistance for subsidy and other hardware costs under the various programmes and projects, the government seem reluctant to provide enough budgets towards technical assistance to mobilize the financial assistance to such programmes and projects, thus hampering the mobilization of subsidy and hardware costs towards the programmes and projects. Under this activity, RERL will provide the much-required technical assistance to AEPC to implement multi-year special programmes as provisioned in the Plans and Programmes of the Government of Nepal and also assist in the implementation of other RE projects implemented by AEPC. With minimal support, it is envisaged that outputs and outcomes of such projects and programmes can be achieved while also developing the capacity of the government, particularly AEPC, to plan, implement and monitor RE projects which ultimately contributes to providing modern energy services to the local people and thereby also increasing the share of renewable energy in the national energy mix of the country.

Among the currently implementing projects/programmes within AEPC, RERL will provide technical assistance to, but not limited to, the government special programme of Himali Solar Mini-Grid Programme, World Bank funded Nepal Mini Grid Project and Solar Pumping Project.

Output 2 – Strengthened policy and institutional capacity to create a conducive environment for renewable energy promotion in Nepal

RERL intends to enhance the participation of financial institutions and the private sector into the RE sector for improved access to financing and overall self-sustaining sectoral growth. For this, RERL will work very closely with commercial banks and the private sector.

Activity 1. Strengthening Central Renewable Energy Fund (CREF):

The Central Renewable Energy Fund (CREF), an intermediate financial mechanism established under the National Rural Renewable Energy Programme (NRREP), has been operational and instrumental in concluding financial closures of large mini hydro and solar mini-grid projects developed with government assistance. However, there are many areas where CREF needs strengthening to carry out this mandate smoothly. RERL is working directly with CREF to design and establish different financial instruments to attract private sector investment in renewable energy projects. With RERL support, CREF has established Soft Credit, Vendor Financing, Credit Guarantee and mechanisms to finance through Local Financial Institutions such as Micro Finances and Cooperatives. However, CREF has been facing multiple operational level challenges such as lack of dedicated credit funds, understaffing, limited capacity building support etc. and has not been able to fully operationalize these instruments. RERL will work closely with CREF to overcome the hurdles in to operationalize these financial instruments. On the other hand, the CREF has smoothly handled GoN subsidies for RE projects as the Handling Bank has dedicated team responsible for the compliance checks and subsidy processing. RERL will thus continue to utilize CREF mechanism to channel financial support to achieve financial closure of RE projects and technical assistance to implement the projects, enhance capacities of developers, communities/beneficiaries, private sector organization, local and provincial governments and other stakeholders involved in the renewable energy sector.

The CREF Business Plan (2017-2021) has developed plans and strategies to mitigate the issues and challenges. In this context, the project aims to continue strengthening of CREF as AEPC is slowly but steadily moving from a subsidy-driven approach to a more market-based approach to project development. The role of CREF is considered very crucial especially in the context that promotion of distributed generation and grid interconnection of smaller RE systems is deemed to play an important part of the future strategy in the renewable energy sector in the country. The major sub activities under this activity are:

- Orientation of banks and financial institutions
- Development of financing mechanisms and its implementation
- Operationalization of credit guarantee mechanisms
- Human resource (expertise) support
- Learning opportunities

Activity 2. Capacity development of Federal, Provincial and Local Governments: The participation of stakeholders is key to implementation and sustainability of RE related initiatives in the country. Under the changed context of federalism, the roles, responsibilities and authorities of various federal, provincial and local level stakeholders have changed. However, the smooth transition of rights and responsibilities along with the capacity to exercise one's authority has been seen to be highly deficient in most cases. RERL, with its vast previous experience on decentralized energy promotion and working with national and local stakeholders including the national and local governments, will prioritize the capacity development of all stakeholders for RE promotion in the country.

Under this activity, the project will take forward capacity development initiatives at all three levels of governments to create a more conducive environment for promotion of renewable energy in Nepal. As part of these efforts, RERL will provide support to AEPC to help it evolve into the 'Centre of Excellence for Renewable Energy'. Support will include preparation of concept notes, business plans, policies, implementation modalities, etc. The project will also work with CREF to help it become a credible financial intermediary channeling subsidies and commercial credit.

Concerned Provincial Governments will be supported to prepare plans for RE development, resource mobilization and allocation, project identification and implementation, supervision, monitoring and evaluation. As the mandate for RE and hydropower projects up to 1MW and related electricity distribution networks are with the local governments, they will be supported to prepare Municipal Energy Plans (MEPs), resource mobilization and allocation, project identification and implementation, supervision, monitoring and evaluation. Both provincial and local governments will be encouraged to invest in RE projects and to create a conducive environment to attract private investments and services in their respective geographical areas. The capacity development activities targeting the local and provincial governments are envisioned to carried out with the help of relevant stakeholders such as Municipal Association of Nepal (MuAN) and National Association of Rural Municipalities in Nepal (NARMIN). RERL will work closely with other UNDP supported projects such as Provincial and Local Government Support Programme (PLGSP) to streamline cooperation, non-duplication and cost effectiveness of capacity development activities at the local and provincial government level. Likewise, The private sector will be supported to both invest and provide services for RE development.

Activity 3. Capacity enhancement of AEPC: In changed context of federalism, the roles and responsibilities of AEPC have changed more towards being a facilitating agency rather than an implementing agency with the implementing roles of RE development and promotion being devolved to the local and provincial governments. With unparalleled vast experience and knowledge within AEPC based on the past two and half decades of active engagement within the RE sector, it is envisaged that this expertise and knowledge could contribute nationally and internationally towards RE promotion and development. Hence, AEPC could be developed as a "Centre of Excellence on Renewable Energy" as part of the repositioning of the organization in the changed federal context. One of the major gaps within AEPC that would facilitate the graduation towards being a knowledge hub nationally and internationally is the lack of adaptive research activities within AEPC. UNDP/RERL will take this opportunity to establish a research wing within AEPC, which could be an important milestone into development of AEPC as a Centre of Excellence on Renewable Energy. There is huge opportunity for renewable energy development within the country and globally, under similar context of Nepal, where AEPC could essentially contribute. The establishment of research wing would reposition AEPC into conducting adaptative researches on areas related to renewable energy that have previously been overlooked mostly due to its core focus on implementation. The research would give an opportunity to critically analyze renewable energy related aspects and deduce inferences that would not only benefit project planning and implementation at the national level but also customize RE development in similar developing countries. The research wing will build AEPC's collaboration with the academic sector thus, being able to mobilize available academic research grants from international

universities and research institutions while also providing opportunities for engagement of young students and professionals into the sector. The establishment of the research wing is expected to contribute towards documentation gaps existing within AEPC. This research activities would also provide an opportunity for AEPC and its personnel as a effective mode of capacity development and wider visibility in the field of renewable energy internationally too. Some of the major sub activities envisaged within this activity are:

- Scoping activities for establishment of research wing within AEPC
- Support towards establishment of a research wing
- Establish and strengthen linkages with national and international academic institutions, private sector, banks and financing institutions and development partners
- Explore opportunities for resource mobilization
- Peer-to-Peer (P2P) learning wherever possible
- Knowledge exchange, experience sharing and technology transfer

Output 3 – Enhanced partnerships for renewable energy development in Nepal

Resource availability for RE development in countries like Nepal are dwindling as Nepal looks to graduate to middle income country soon. Similarly, cooperation and collaboration among national and international agencies is necessary to plug technical and financial resources gaps that ultimately aims to ensure sustainability of interventions while optimizing the available resources and expertise. Partnership remains UNDP's core philosophy of project design and implementation especially in energy projects. RERL, recognizing the national need to mobilize resources and foster partnerships in the RE sector, will assist both UNDP and AEPC to develop partnerships and potentially mobilize resources from different available sources.

Activity 1. Collaboration with partners: RERL will coordinate and collaborate with other national and international partners and projects to contributing towards the larger expansion and adoption of RE systems in Nepal. The Renewable Energy and Energy Efficiency Capability for the Hindu Kush Himalaya (REEECH), a project under International Centre for Integrate Mountain Development (ICIMOD), works to expand the adoption of green energy solutions among HKH communities and enterprises. RERL will seek to further strengthen partnership with ICIMOD so that the strengths of ICIMOD, AEPC and UNDP could be synchronized for the greater benefit of the sector within the country. ICIMOD has proven capacity in data management, technology transfer and conducting trainings in the sector. RERL, on behalf of AEPC, will look to collaborate and strengthen these capacities thereby assisting AEPC in plugging the existing gaps within AEPC. This synergy while strengthening the partnership among like-minded organizations will also come up avenues for other partners to collaborate based on strengths and needs of the sector. UNDP/RERL will work towards fostering partnerships with national and international partners for policy inputs for sectoral growth and expansion that bolsters the capacity of the Government of Nepal to plan and implement RE projects for sustainable livelihood of the people, especially the vulnerable and underprivileged.

Activity 2. Resource Mobilization: UNDP, through the RERL project, aims to design and develop other RE projects in close collaboration with AEPC mainly focusing on technology promotion, capacity development, promotion of livelihood options, empowering women and vulnerable groups and sustainability of RE projects with resource mobilization from national and external partners and donors. RERL has been supporting both AEPC and UNDP to prepare concepts and projects for resource mobilization. RERL will continue to support both the organizations in preparing necessary documentations especially for Green Climate Fund (GCF). Concepts regarding enhancing operation of micro hydro plants, energy for agriculture, electric mobility and energy efficiency will be further strengthened and developed with close coordination with AEPC. Furthermore, the project will explore possibilities of resource mobilization for RE energy promotion in the country with development partners working in the sector including, but not limited to, GCF, Town Development Fund, Global Environment Facility (GEF), Norway, Denmark, European Commission, UK and ICIMOD.

Resources Required to Achieve the Expected Results

Renewable Energy for Rural Livelihood (RERL)
April 2020-December 2024
Cost Estimate

HEADING	Amount for TA (in Million USD)				Project Investment (in million USD)	REMARKS
	Funded		Unfunded			
	UNDP	Others	UNDP	Others		
Output 1: Improved access to clean, affordable and reliable energy solutions						
Activity 1. TA for commissioning and sustainable operation of mini hydro and solar mini grid projects under SASEC off-grid component	0.962	1.5	-	1	16.5	ADB loan and grant
Activity 2: Increase utilization of energy from micro hydro systems in remote rural off-grid areas	-	1.235	0.48	-	1.235	Government of Japan
Distributed generation – MHP interconnection	-	-	0.5	-	2	Loan and Equity
Activity 3. Improve access to universal modern energy services for vulnerable groups through solar energy solutions	-	1.085	0.625	-	1.085	Government of Japan
Activity 4. TA to AEPC for implementation of its projects and programmes	-	-	0.232	-	2	Climate proofing, PEU, DFS/DED
Output 2: Strengthened policy and institutional capacity to create a conducive environment for RE promotion in Nepal						
Activity 1. Strengthening of CREF	-	-	0.125	-	2.5	
Activity 2. Capacity development of LGs/PGs	-	-	0.165	0.16	3.44	
Activity 3. Capacity development of AEPC			0.15	-	-	HR
Output 3: Enhanced partnership for renewable energy development in Nepal						
Activity 1. Collaboration with partners	-	-	0.05	-	1	
Activity 2. Resource mobilization	-	-	0.05	0.925	-	GCF, GEF, ICIMOD
Operation support costs	0.198	-	0.172	0.452	-	
Total estimated budget	**1.16**	**3.82**	**2.5**	**2.2**	**29.76**	

Partnerships

RERL is mainly providing technical assistance to AEPC to install 4.3MW mini hydro and 0.5MW solar mini grids and productive uses of energy also ensuring achievement of GESI targets. Whereas the subsidies as well as credit necessary to achieve financial closure of the projects comes from SASEC/ADB and AEPC, thus there is inherent complementarity and supplementarily among the main partners. This partnership between AEPC, ADB and UNDP is a continuation of the partnership during the GEF RERL period, where GEF RERL provided TA to implement SASEC mini hydro and solar mini grids projects. Partnership works best when the partners complement and supplement each other which is demonstrated by the partnership between GEF RERL and the ADB/AEPC/SASEC.

Continuing the partnership between GEF RERL and Renewable Energy for Rural Areas (RERA) project of AEPC and GiZ, RERL will engage in updating the methodology for preparation of MEPs and digitalization and 'commercial operation' of MHPs. RERL will seek opportunities to work with National Renewable Energy Project (NREP) of AEPC and FCDO UK. There are several areas of work, including capacity building of provincial and local governments and preparation of energy plans, aggregation of potential projects for financing, engineering design and construction supervision of RE projects, promotion of PEU and commercial operation of RE projects, where RERL's experience can complement NREP's activities.

As local governments and rural people without access to electricity are the main beneficiaries of the project, they are by nature also key partners. They are responsible for equity collection, loan, land acquisition, local conflict resolution, roles and responsibilities which are vital for successful project implementation. Further, during the post installation period, as they are responsible for smooth operation and management, their roles are vital for promotion of productive end uses and ensuring that women and marginalized communities also benefit from access to electricity.

RERL will endeavor to establish partnerships and collaboration with like-minded national and international organizations to share costs, knowledge and experiences as a key strategy for synergetic environment to support the rural people.

Risks and Assumptions

Even though project implementation presents substantial challenge, the proposed project is feasible. The overall project risk is moderate. Major areas of risk include the external investments and political environment, the response of the private sector and banks to the opportunity to invest in off-grid RE electrification projects, timely equity collection by developers and local governments, effectiveness of project management and stakeholder coordination. Acquisition of land required for civil and other structures done by the developer following Environment and Social Safeguard Framework of ADB on time and without conflicts is necessary for timely completion of projects. Further, both the financial and managerial capacities of vendor/supplier/contractor to execute their responsibilities on time is also crucial for smooth implementation of the projects and avoidance of cost overruns.

Key assumptions include banks and private sector being interested in financing commercially viable off-grid RE power projects, the investment environment and the local and national political environments remain conducive to project implementation, the project is driven by strong leadership from the National Project Manager (NPM) and strong support of the National Project Director (NPD), there are potential for productive energy uses and local entrepreneurs are interested in establishing productive application enterprises and possess the capacity to benefit from project efforts to support them in enterprise development and achievement of market linkages and CREF funds are available to support capacity targets.

However, it is worth noting that there exists certain risks and assumptions on the preparedness of projects for implementation including timely resolution of social and technical issues, political disturbances, timely financial closure especially equity collection and loan disbursement, procurement, land acquisition, addressing of unforeseen variations due to geological conditions, landslides, floods, contractors' capability, efficiency and effectiveness for procurement and supply of goods/equipment, transportation and timely work completion.

For implementation of the SASEC project, RERL will engage closely with AEPC, SASEC, ADB, local governments and the beneficiary communities to plan and implement the projects in an efficient and coordinated manner. RERL will ensure management, oversight and monitoring of implementation of ESS during the implementation of the projects.

Key assumptions beyond the control of the project are related to some of the risks mentioned above. They include: (1) The banks and private sector are interested in financing commercially viable off-grid RE power projects; (2) The investment environment and the local and national political environments remain conducive to project implementation; (3) The project will be driven forward towards the objectives by strong leadership from the NPM and strong support of the NPD; (4) Local people are interested in productive applications and possess the capacity to benefit from project efforts to support them in enterprise development and achievement of market linkages; (5) AEPC remains supportive of the project; and (6) CREF funds are available to support capacity targets.

Stakeholder Engagement

People without access to electricity living in remote rural areas and their local governments are the main beneficiaries of the project and by nature the key stakeholders. They are responsible for equity collection, loan, land acquisition, local conflict resolution, etc., roles and responsibilities that are vital for project implementation. Further, during the post installation period, as they are responsible for smooth operation and management, their role for promotion of productive end uses and ensuring that women and marginalized communities also benefit from access to electricity are important. The project will support AEPC to sign Memorandum of Understandings (MoUs) with local governments to both oversee project implementation as well as invest in project development. RERL will also support these local governments to prepare and execute their respective MEPs.

The beneficiary communities will be mobilized following the REDP/RERL Community Mobilization Guidelines and supported to establish legal institutions, such as cooperatives and companies which act as the SPVs, who will be responsible for project development, implementation, operation and management.

The AEPC, under MoEWRI, is the Implementing Partner and is the key governmental stakeholder. Besides, MoEWRI which is responsible for the power sector of the entire country and the Ministry of Finance (MoF) are other governmental partners of the project and are represented in Project Executive Board (PEB) responsible for overseeing project implementation. Other governmental agencies or authorities related to RE and electricity in Nepal include the followings:

- **Ministry of Finance (MoF):** Responsible for allocation of national resources, management of public expenditure, mobilization of both internal and external resources, and performance of public investments. All government allocations and donor assistance for the off grid RE sector are managed by this Ministry.

- **Ministry of Energy, Water Resources and Irrigation (MoEWRI):** Responsible for the power sector of the entire country, oversees the development and management of both national grid-connected electricity systems and off-grid projects. The Ministry is also responsible for utilization and management of water resources of the country, including for electricity production and management.
- **Ministry of Federal Affairs and General Administration (MoFAGA):** Responsible for coordination between the federal government and provincial and local governments. It is also responsible for strengthening the capacity of provincial and local governments to enhance decentralized development.
- **Nepal Electricity Authority (NEA):** NEA is the government utility responsible for planning, construction, operation, maintenance of electricity generation, transmission, and distribution facilities in Nepal's power system. It also purchases power from private developers and exchanges power with India to meet the nation's needs. In the past, NEA has installed some off-grid systems, but these operate at substantial loss. As a result, new ones are no longer being pursued. Some of the systems built are either not operating or have been leased out to private companies or communities. NEA is currently showing interest in continuing operation of off-grid systems at remote locations and interconnecting isolated systems to the grid.
- **Provincial Governments:** The Constitution and the Local Government Operation Act 2017 give conditional mandate on RE development to federal, provincial and local governments. Though the roles and responsibilities of the provincial governments in relation to RE and power sector development is yet to be fully clarified, some of them have been engaged in supporting local governments to complete/rehabilitate micro hydropower projects and design solar PV projects and indicated interest in investing in RE projects.
- **Local Governments:** The local governments (municipalities and rural municipalities) will support in selection of project beneficiaries and interventions, share the cost of establishing the schemes and take responsibility of sustainable operation and management of the schemes beyond the project period.
- **Private Sector:** The role of the private sector in RE in Nepal so far has been limited to mostly supply and installation of systems. Very few off-grid projects have been developed with private investment. GEF RERL and AEPC/CREF designed and operationalized a number of financial instruments to lower the risk of private investment in off-grid RE projects and two mini hydro and a few solar PV projects have been developed by private companies. RERL will continue to work with CREF to attract private investment in RE projects.

South-South and Triangular Cooperation (SSC/TrC)

RERL has submitted a project proposal for Economic Empowerment of Women through Energy Access in Nepal through UNDP to China South-South Cooperation Assistance Fund (CSSCAF) for funding. Likewise, RERL has supported AEPC to prepare a concept note and a draft MoU for collaboration on solar PV with Asia-Pacific Researcher and Training Centre for Solar Energy, China. If these proposals materialize, AEPC will have substantial engagement with China. Further, RERL sees an opportunity to work with Renewable Energy and Energy Efficiency Centre for Hindu Kush Himalaya Region (REEECH) of ICIMOD to share its experiences and learn from neighboring countries with similar socio-economic conditions.

Digital Solutions[1]

The project, based on its vast experiences of promoting renewable energy, will use digital solutions to build on its initiatives to attain sustainability of energy systems and enhance transparency and accountability of governance and management models for ensuring quality and reliable energy supply to the beneficiary households. While technology transfer and wider dissemination and adoption of newer innovations on the technological front will be the focus of the project, increased use of digital solutions to strengthen existing practices will also be pursued such as development and adoption of accounting software for project financial management, development of knowledge products such as documentaries, short videos, etc., development of mobile applications for project identification, planning and monitoring.

Knowledge

The project team will prepare the Project Completion Report summarizing the results achieved, lessons learned, problems faced and areas where results may not have been achieved. It will also lay out recommendations for any

[1] *Please see the Guideline "Embedding Digital in Project Design".*

further steps that may need to be taken to ensure sustainability and reliability of project's results. Results and lessons learned from the project will be disseminated through existing information sharing networks and forums like national knowledge management institutions, national and international workshops and seminars.

The project will participate, as relevant and appropriate, in scientific, policy-based and/or any other networks, which may benefit project implementation through lessons learned. The project will identify, analyze, and share lessons learned that might be beneficial in the design and implementation of similar projects in future. There will be a two-way information flow between this project and other projects of similar focus nationally and internationally. Experiences will be shared in the region as relevant.

Similarly, the project will prioritize experience sharing and peer-to-peer learning at a larger scale with participation in international forums and seminar and conferences. The project will disseminate information through various print and digital media including preparation of documentaries, creation and regular updating of websites, production of success stories that will be shared to a larger audiences.

Sustainability and Scaling Up

UNDP has been closely working with AEPC since its establishment in 1996 to develop renewable energy in Nepal. In 1996, UNDP and the Government of Nepal launched Rural Energy Development Programme (REDP) which was operationally closed in 2011 and was followed by RERL. These two flagship projects of UNDP and AEPC demonstrated many innovative modalities for promotion of the sector, particularly micro hydro. REDP's decentralized energy development modality giving key role to the erst-while District Development Committees was later incorporated in the Rural Energy Policy 2006 and Energy and Environment Units were established in all 75 districts. This indicates full ownership of the concerned governmental agencies to internalize and scaleup effective ideas and mechanisms. Likewise, the community developed, owned and managed micro hydro modality introduced by REDP was later adopted by other AEPC programmes and replicated all over the country. There are now more than 3,000 AEPC supported micro hydropower projects all over Nepal providing electricity access to over half million rural households.

Furthermore, in 2019, AEPC and the World Bank have already initiated implementation of private sector led Nepal Mini Grid Project to promote mini/micro hydro and solar mini grids through commercial model. This project also has incorporated many lessons from GEF RERL.

The endeavor to promote mini hydro and large solar PV systems with private investment pioneered by GEF RERL in Nepal was instrumental in ADB and AEPC's decision to launch SASEC Off-grid Component to promote mini hydro and solar mini grids. RERL will work with AEPC to mobilize resources to replicate the lessons from implementation of SASEC. As the local governments are given key responsibility to invest in project development and oversee implementation, their capacity will be enhanced along the way and they will be able to sustainably operate these projects. Further, transparent and accountable governance and incentive-based management system in each RE project will ensure sustainable operation of these projects.

IV. PROJECT MANAGEMENT

Cost Efficiency and Effectiveness

Though over 95% of the population has access to electricity, only 93% is reached by the national grid. To provide affordable, reliable, sustainable and modern energy services to the remaining population, which overwhelmingly lives in sparse and scattered settlements in the remote mountains, off-grid solutions are the least costly options. Depending on the number of households in a settlement, distance between the settlements, energy requirements and demands, off-grid solutions like mini/micro hydro or solar mini grids or solar home systems are the 'Best Available Technology' for electricity access. In case of demand for daytime load, solar PV or mini/micro hydro depending on the demand size and location is the cheapest option. Likewise, if electricity is required only for lighting and operating small appliances, solar home system may be the best option. In the methodology developed for municipal energy planning, the best available option for rural electrification is identified on the basis of levelized cost of electricity.

AEPC will work for coordination and cooperation among its programmes and projects like RERL, RERA, NREP, NMGP and SREP to create synergy avoiding duplications and complementing/supplementing each other. RERL will engage with AEPC in identification, feasibility studies, detailed engineering design, construction supervision, productive end

use promotion and commercial operation, community mobilization, institution establishment and strengthening, project financing, etc. wherever and whenever possible.

Project Management

Mini Hydro: RERL has provided technical assistance to complete four mini hydropower projects. As there are more projects either under construction or at different stages of development, RERL will ensure smooth completion of such projects and smooth transition from each stage to the other through proper planning and implementation. A Site Coordinator, who will oversee the construction for timely and efficient completion of the projects, will be stationed at each project. A Field Coordinator for community mobilization, institution establishment and strengthening, facilitation for equity collection, conflict resolution, GESI mainstreaming and productive energy use promotion will also be stationed at the project site. Depending on the capacity and complexities, the Coordinator may be supported by an Overseer. The projects with their capacities and beneficiary households are given in the Table 1 below.

Table 1: Location of Mini Hydropower Projects

S.N.	Projects	District	kW	HHs#	Status
1.	Simrutu Khola	Rukum (W)	200	1386	Completed
2.	Giri Khola	Jumla	200	1840	Completed
3.	Middle Phawa Khola	Taplejung	500	2070	Completed
4.	Khatyad Khola	Mugu	500	2564	Under construction
5.	Lower Bom Khola	Solukhumbu	200	617	Completed
6.	Patrasi Chukeni Khola	Jumla	998	6250	Under construction
7.	Sani Bheri Khola	Rukum (E)	998	5039	Under construction
8.	Ankhe Khola	Dolpa	750	3018	Pipeline
Total			**4,346**	**22,784**	

#Total expected connections

Solar Mini Grid: The installation of all 9 solar mini grids has been completed by mid-2021. The solar mini grids' locations, their capacities and number of beneficiary households are given in the Table 2 below.

Table 2: Location of Solar Mini Grids

S.N.	Name of Subprojects	District	kW	HHs	Status
1.	Chisapani SMG	Sindhuli	35	90	Completed
2.	Ramitekhola SMG	Morang	30	75	Completed
3.	Olane SMG	Panchthar	25	70	Completed
4.	Saptami SMG	Panchthar	70	110	Completed
5.	Gutu SMG	Surkhet	100	344	Completed
6.	Sugarkhal SMG	Kailali	75	215	Completed
7.	Dandapur (Malladehi) SMG	Baitadi	30	110	Completed
8.	Hilepani SMG	Okhaldhunga	50	233	Completed
9.	Thabang SMG	Rolpa	150	365	Completed
Total			**565**	**1,612**	

Asset transfer from old RERL project: UNDP has implemented transitional RERL project from July 2019 to March 2020. All the assets including office equipment and vehicles being utilized by transitional RERL project will be transferred to the new RERL project.

Audit arrangements: The Government will provide UNDP Resident Representative with certified periodic financial statements, and with an annual audit of the financial statements relating to the status of UNDP funds according to the established procedures set out in the programming and finance manuals. The audit will be conducted according to UNDP financial regulations, rules and audit policies by the legally recognized auditor of the Government, or by a commercial auditor engaged by the Government.

UNDP Support Services: UNDP Nepal will provide all required support in terms of participating in various Board and advisory meetings as required and by promoting the project in national and donor community.

V. RESULTS FRAMEWORK[2]

Intended Outcome as stated in the UNSDCF/Country Programme Results and Resource Framework: **UNSDCF/CPD Outcome 3:** By 2027, more people, especially women, youth and the most marginalized and poor, increasingly benefit from and contribute to building an inclusive, sustainable, climate-resilient and green society and reduced impacts of disasters at federal, provincial, and local levels.
Outcome indicators as stated in the Country Programme Results and Resources Framework, including baseline and targets: **CPD Output 3.2:** Ratio of renewable energy in total energy consumption (%) Baseline (2019): 5%; Target (2025): 37.3%
Applicable Output(s) from the UNDP Strategic Plan: Resilience built to respond to systemic uncertainty and risk. **Output 3.1 :** Green, resilient and inclusive development planned, financed and implemented at federal, provincial and local levels Output 3.4. Reliable, affordable, and efficient clean energy solutions promoted
Project title and Atlas Project Number: Renewal Energy for Rural Livelihood (RERL); **Project ID:** 00117173

EXPECTED OUTPUTS	OUTPUT INDICATORS[3]	DATA SOURCE	BASELINE		TARGETS (by frequency of data collection)						DATA COLLECTION AND VERFICATION METHODS & RISKS
			Value	Year	2020	2021	2022	2023	2024	FINAL	
Output 1 Improved access to clean, affordable and reliable energy solutions	MW of mini/micro hydro installed/rehabilitated	AEPC, SASEC & RERL	0.98 MW	2019	0.2 MW	0.4 MW	2.0 MW	1.0 MW	1.75 MW	*5.35 MW*	Periodic reports, completion report, RERL MIS
	kW of solar mini grid installed	AEPC, SASEC & RERL	420 kW	2019	150 kW	0	0	0	0	150 kW	Periodic reports, completion report, RERL MIS
	Percentage of power capacity for economic activities	Project MIS	0	2019						20%	RERL MIS, Business Assessment/Plan, field visit
	No. of households adopting electric cooking	Project MIS	10	2019						50000	Periodic reports, RERL MIS, field visits, surveys

[2] UNDP publishes its project information (indicators, baselines, targets and results) to meet the International Aid Transparency Initiative (IATI) standards. Make sure that indicators are S.M.A.R.T. (Specific, Measurable, Attainable, Relevant and Time-bound), provide accurate baselines and targets underpinned by reliable evidence and data, and avoid acronyms so that external audience clearly understand the results of the project.

[3] It is recommended that projects use output indicators from the Strategic Plan IRRF, as relevant, in addition to project-specific results indicators. Indicators should be disaggregated by sex or for other targeted groups where relevant.

EXPECTED OUTPUTS	OUTPUT INDICATORS[3]	DATA SOURCE	BASELINE		TARGETS (by frequency of data collection)						DATA COLLECTION AND VERFICATION METHODS & RISKS
			Value	Year	2020	2021	2022	2023	2024	FINAL	
	No. of micro hydro projects rehabilitated for climate resilience	Project MIS	104	2019	0	0	10	20	20	50	Periodic reports, RERL MIS, completion reports
Output 2 Strengthened policy and institutional capacity to create a conducive environment for RE promotion in Nepal	2.1: No of financial closure of RE projects	Project MIS	4	2019						16	Periodic reports
	2.2: No of Municipal and Provincial Energy Plans prepared	Project MIS	7	2019						25	Periodic reports, RERL MIS, Energy Plans
	2.3: No. of national RE policy documents drafted	Project MIS	8	2019		1	2	1		4	Policy documents, periodic reports, RERL MIS
Output 3 Enhanced partnerships for renewable energy development in Nepal	3.1: No. of project proposals/concept notes developed and submitted to different agencies for resource mobilization	Project MIS	28	2019	2	5	10	8	5	30	Concept notes, periodic reports, staff performance appraisal reports

VI. MONITORING AND EVALUATION

In accordance with UNDP's programming policies and procedures, the project will be monitored through the following monitoring and evaluation plans:

Monitoring Activity	Purpose	Frequency	Expected Action	Partners (if joint)	Cost (if any)
Track results progress	Progress data against the results indicators in the RRF will be collected and analyzed to assess project progress in achieving the agreed outputs.	Quarterly	Slower than expected progress will be addressed by project management.	AEPC, ADB	
Monitor and Manage Risk	Identify specific risks that may threaten achievement of intended results. Identify and monitor risk management actions using a risk log. This includes monitoring measures and plans that may have been required as per UNDP's Social and Environmental Standards. Audits will be conducted in accordance with UNDP's audit policy to manage financial risk.	Quarterly	Risks are identified by project management and actions are taken to manage risk. The risk log is actively maintained to keep track of identified risks and actions taken.	AEPC, ADB	
Learn	Knowledge, good practices and lessons will be captured regularly, as well as actively sourced from other projects and partners and integrated back into the project.	Annually	Relevant lessons are captured by the project team and used to inform management decisions.	AEPC, ADB	
Annual Project Quality Assurance	The quality of the project will be assessed against UNDP's quality standards to identify project strengths and weaknesses and to inform management decision making to improve the project.	Annually	Areas of strength and weakness will be reviewed by project management and used to inform decisions to improve project performance.	GoN, AEPC, ADB,	
Review and Make Course Corrections	Internal review of data and evidence from all monitoring actions to inform decision making.	Annually	Performance data, risks, lessons and quality will be discussed by the project board and used to make course corrections.	GoN, AEPC, ADB	
Project Report	Progress report will be presented to the Project Board and key stakeholders, consisting of progress data showing the results achieved against pre-defined annual targets at the output level, the annual project quality rating summary, an updated risk log with mitigation measures, and any review reports prepared over the period.	Annually, and at the end of the project	CO will provide quality assurance of the Project Reports.	GoN, AEPC, ADB	
Project Review (Project Board)	The project's board will hold regular project reviews to assess the performance of the project and review the Multi-Year Work Plan to ensure realistic budgeting over the life of the project. In the project's final year, the Project Board shall hold an end-of project review to capture lessons learned and discuss opportunities for scaling up and to socialize project results and lessons learned with relevant audiences.	Quarterly	Any quality concerns or slower than expected progress should be discussed by the project board and management actions agreed to address the issues identified.	GoN, AEPC, ADB	

Evaluation Plan

Evaluation Title	Partners (if joint)	Related Strategic Plan Output	UNSDCF/CPD Outcome	Planned Completion Date	Key Evaluation Stakeholders	Cost and Source of Funding
Terminal Evaluation	ADB/GoJ	Output 5.1; Output 5.2	CPD Outcome 1 &3	Q4 2024	Ministry, AEPC, UNDP, ADB, GoJ	30,000 (UNDP TRAC)

VII. MULTI-YEAR WORK PLAN

EXPECTED OUTPUTS	PLANNED ACTIVITIES	Planned Budget by Year					RESPONSIBLE PARTY	PLANNED BUDGET		
		2020	2021	2022	2023	2024		Source	Description	Amount
Output 1: Improved access to clean, affordable and reliable energy solutions *(GEN: 2)*	1.1 TA for commissioning and operation of projects under SASEC	637,258	488,387	374,355	-	-	AEPC/SASEC	ADB	Programme	1,500,000
		-	-	-	450,645	364,170		Donor	Programme	814,815
		269,485	354,704	87,981	100,000	150,000		UNDP	Support	962,170
	1.2 Increase utilization of energy from micro hydro systems in rural areas	-	-	-	734,879	500,121	UNDP	GoJ[4]	Grant	1,235,000
		50,000	50,000	80,000	150,000	150,000		UNDP	TA	480,000
	1.3 Promotion of distributed generation	-	50,000	250,000	50,000	150,000	AEPC	UNDP	TA	500,000
	1.4 Improve access to modern energy services through solar solutions	-	-	-	605,195	479,805	UNDP	GoJ	Grant	1,085,000
		150,000	300,000	75,000	-	100,000		UNDP	TA	625,000
	1.5 TA to AEPC for implementation of its projects and programmes	-	-	-	100,000	132,170	AEPC	UNDP	TA	232,170
	Sub-Total for Output 1									**7,434,155**
Output 2: Strengthened policy and institutional capacity to create a conducive environment for RE promotion in Nepal *(GEN: 2)*	2.1 Strengthening of CREF	-	25,000	30,000	30,000	40,000	AEPC	UNDP	TA	125,000
	2.2 Capacity development of LGs/PGs	-	-	25,000	75,000	60,185	AEPC	Donor	TA	160,185
		25,000	25,000	25,000	50,000	40,000		UNDP	TA	165,000
	2.3 Capacity development of AEPC	-	-	50,000	50,000	50,000	AEPC	UNDP	TA	150,000
	Sub-Total for Output 2									**600,185**
Output 3: Enhanced partnerships for renewable energy development in Nepal *(GEN: 2)*	3.1 Collaboration with partners	-	-	20,000	20,000	10,000	AEPC	UNDP	TA	50,000
	3.2 Resource mobilization	-	-	25,000	385,000	362,963	UNDP	Donor	TA	925,926
					25,000	25,000		UNDP	TA	50,000
	Sub-Total for Output 3									**1,025,926**

[4] GoJ – Government of Japan

Operation Cost[5#]	OPERATION	50,000	30,000	130,000	57,830	102,830	UNDP	UNDP	TA	**370,660**
Project Management Support	Project Technical Staff cost, Supplies, Communication etc.	21,381	39,531	156,911	84,214	150,000	UNDP	Donor	GOE service cost	**452,037**
	Sub-Total for Output (Programme Management Support)									**822,697**
	Grand TOTAL									**9,882,963**

Multi-year Budget Summary

Donor	2020	2021	2022	2023	2024	Total
ADB	637,258	488,387	374,355	-	-	**1,500,000**
Government of Japan	-	-	-	1,340,074	979,926	**2,320,000**
UNDP	269,485	354,704	87,981	207,830	240,000	**1,160,000**
	TOTAL Funded Amount					**4,980,000**
UNFUNDED BUDGET						
Donor	**2020**	**2021**	**2022**	**2023**	**2024**	**Total**
Donor	21,381	39,531	206,911	994,859	937,318	2,200,000
UNDP	275,000	480,000	610,000	425,000	710,000	2,500,000
Government	762,000	762,000	762,000	1,016,000	1,778,000	5,080,000
	TOTAL Unfunded Amount					**9,780,000**

[5] Please Note: The evaluation cost of the project is included in the operation cost of the project

VIII. GOVERNANCE AND MANAGEMENT ARRANGEMENTS

RERL is implemented by the AEPC, under the MoEWRI, following the NIM Guidelines as agreed between GoN and UNDP. While the management structure of RERL will be guided by the provisions agreed with the government and UNDP procedures as governed by NIM guidelines, the overall management structure will fit into AEPC structure to avoid duplications during implementation. The management arrangement for RERL is discussed below.

Government Cooperating Agency: The MoEWRI, as the cooperating agency, is responsible for high-level monitoring of the project on behalf of the Government of Nepal, promote initiatives undertaken by the project nationally as best practice case, and ensure appropriateness of interventions in meeting national priorities. The MoEWRI may coordinate with other relevant ministries and agencies in order to provide inputs to the project as and when needed.

- The GoN will cooperate and assist the project personnel in the carrying out the Services.
- The GoN will provide free of charges the services, facilities, equipment, documents and information as may be required in the course of the project.

Project Executive Board: To facilitate smooth and effective implementation of project activities, PEB for the current phase of RERL will be reconstituted to reflect the new stakeholders of the project. The PEB is responsible for taking management decisions with respect to project implementation and management, its three main roles are described below.

- The PEB will have representation of MoEWRI, MoF, ADB, Nepal Mini/Micro Hydropower Development Association (NMHDA) representing the Private Sector and National Association of Rural Municipalities in Nepal (NARMIN) representing the local governments as **Senior Beneficiaries** of the project. The Senior Beneficiaries' primary function within the Board is to ensure realization of project results from the perspective of project beneficiaries.
- UNDP Nepal is the **Senior Supplier** and its function within the Board is to provide guidance regarding the technical feasibility of the project.
- The National Project Director will be the **Chair** of the PEB and is responsible to manage the project and ascertain timely project outputs.

The PEB plays a critical role in project monitoring and evaluation, assuring quality and performance improvement. The PEB ensures that required resources are committed and arbitrates on any conflicts within the project or negotiates a solution to any problems with external bodies. In addition, it approves appointment and responsibilities of the National Project Manager (NPM) and any delegation of its project assurance responsibilities. The RERL PEB will meet every quarter following the Gregorian Calendar and approve Annual Work Plan (AWP), Quarterly Work Plan (QWP) and any essential deviations from the original plans.

National Project Director (NPD): The Executive Director of the AEPC serves as the NPD of RERL with the responsibility for overseeing overall project implementation and ensuring that the project objective and outcomes are achieved in a timely and cost-effective manner. The NPD is assisted by a full-time NPM. The NPD is responsible for coordinating and maintaining synergy in achievement of results, exchange of knowledge and expertise from RERL to AEPC and vice-versa. The NPD provides guidance to the NPM on both strategic and project implementation issues.

Project Assurance: PEB is supported by UNDP Portfolio Manager to assure that the project is implemented as per its objectives by providing oversight and monitoring functions independently.

National Programme Manager (NPM): The NPM is responsible for day-to-day management, administration and coordination for timely project implementation. The main functions of the NPM is to develop project work plans, implement them in line with the project objectives, strategies and result frameworks, and prepare reports as required by the GoN, UNDP and other funding agencies like ADB. The NPM is also responsible for monitoring work progress and ensure timely delivery of outputs in a cost-effective manner as per the plan. The NPM will ensure high quality of planning, management, implementation, technical/financial compliance, reporting and monitoring.

Technical Experts/Professional staff: As RERL will support to achieve the specific objectives of SASEC project and broader objectives of AEPC and UNDP, it will have the following professional staff:

- **Senior National Advisor (SNA):** Apart from the responsibilities of NPM of the TA component, the SNA will be responsible for maintaining internal and external relations with stakeholders including, but not limited to, AEPC, CREF, ADB and UNDP in order to achieve better synergy and coordination. Additionally, the SNA will work towards institutionalizing mini hydro and solar mini grid projects into the national policy regimes while also assisting AEPC, ADB and UNDP for RE development activities in Nepal.
- **Mini Hydro Expert (MHE):** The MHE will be responsible for overall construction supervision of mini hydro projects. The MHE will work closely with, provide guidance to and monitor performance of Site Coordinators, Mini Hydro Engineer and Overseers for successful and timely completion of mini hydro subprojects.
- **Solar Power Expert (SPE):** The SPE will be responsible for all electrical components of projects including design of transmission and distribution lines, electrical machines, testing commissioning of mini hydro projects as per the available guidelines. The SPE will also support AEPC and SASEC on grid interconnection of subprojects and other power system related works. The SPE will also supervise Electrical and Solar Engineers.
- **Livelihood Expert (LE):** The LE will be responsible for overall community mobilization and institutional development and strengthening activities of the project. The LE will be responsible for livelihood enhancement and promotion of PEU activities and ensuring that targets pertaining to enterprise development are met. The LE will monitor and supervise the works of Field Facilitators stationed at individual projects.
- **Policy and Institutional Strengthening Expert (PISE):** The PISE will assist the project in coordinated planning among various stakeholders for timely delivery of targets. The PISE will also act as the GESI Focal Point of the project and will oversee the implementation of GESI related activities to meet the agreed GESI targets. The PISE will work closely with GESI teams at AEPC, ADB and UNDP and provide inputs to the project at the national and local level for GESI mainstreaming into the entire project cycle. The PISE will also be responsible for GESI monitoring and reporting of the project.
- **Administration and Finance Officer (AFO):** The AFO will be responsible for ensuring smooth administrative, human resource and financial management of the project. The AFO will provide inputs to project personnel on all matters pertaining to administrative and financial functions during the implementation of project activities including planning and budgeting, procurement of goods and services, auditing, etc.
- **Knowledge Management Officer (KMO):** The KMO will be responsible for quality control and regular monitoring of project interventions. The KMO will also work closely with AEPC and UNDP in generating knowledge products of the project and sharing information to the larger stakeholders of the project. The KMO will also be responsible for establishing appropriate monitoring systems and database management of the project. The KMO will generate various periodic reports required by AEPC, ADB and UNDP.
- **Environment and Social Safeguard Expert (ESSE):** The ESSE will ensure the implementation of environment and social safeguard activities in line with available applicable frameworks. The ESSE shall be responsible for management, oversight and monitoring of implementation of environment and social safeguard compliance of the project during the project preparation and implementation phase.

To support on financial management and compliance, reporting, administrative/logistics activities and transportation 6 support staff (Administration and Finance Associate, Programme Associate and 4 Drivers) working with RERL will continue to provide its services. This will be funded by UNDP Nepal.

For smooth project implementation, both professional and support staff will be retained for the entire project period. Based on requirement, RERL will recruit short term staff to support project implementation. All experts recruited by RERL are supervised and guided by the NPD/NPM.

GESI Focal Point: UNDP has been supporting AEPC to mainstream GESI initiatives through past and on-going projects. As a result, capacity has been built within AEPC to mainstream GESI into overall AEPC programming and implementation. AEPC GESI Manager will coordinate with the GESI focal point of RERL to mainstream GESI initiatives into the project. GESI focal point will collaborate closely with UNDP Gender team during project implementation.

The organizational setup of UNDP/RERL as described above is depicted in the figure below.

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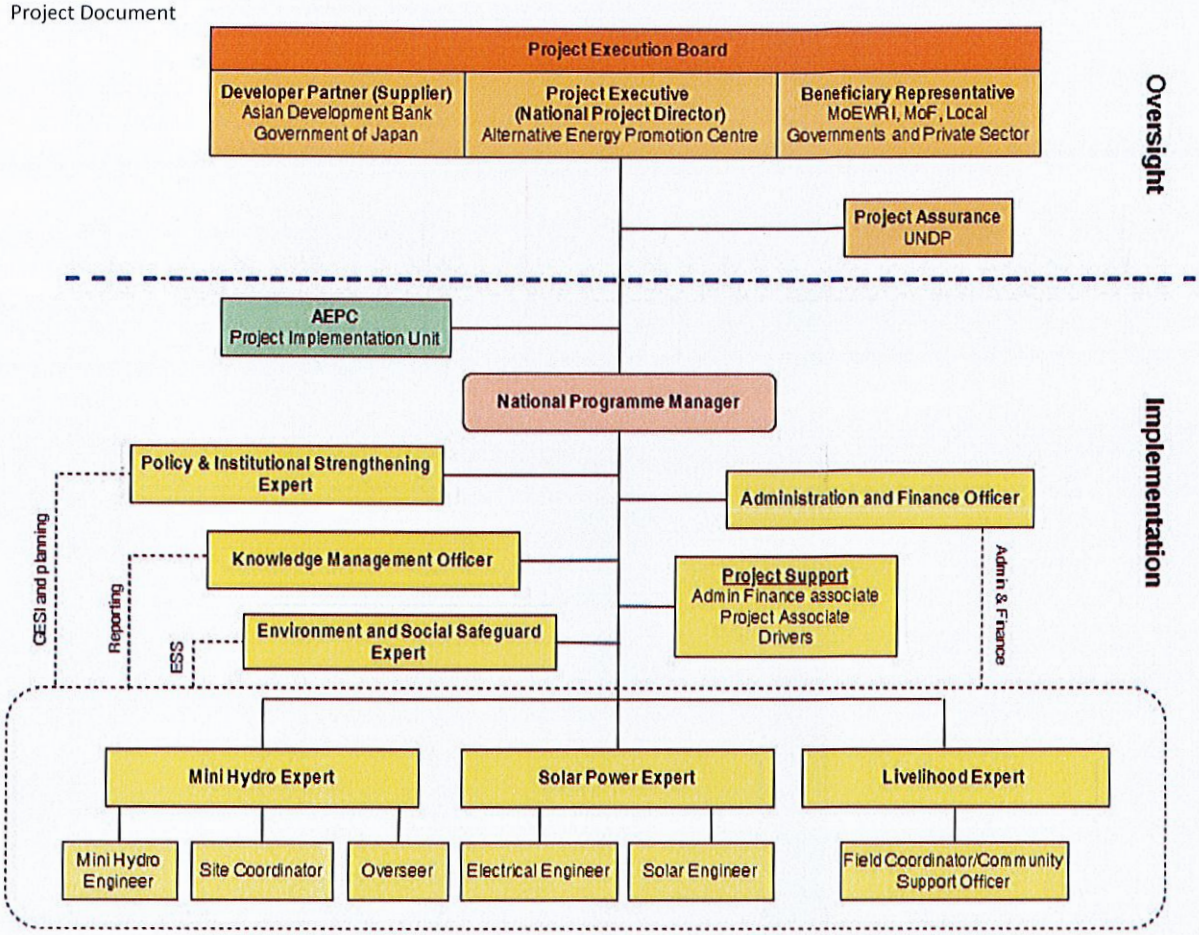

IX. LEGAL CONTEXT

This project document shall be the instrument referred to as such in Article 1 of the Standard Basic Assistance Agreement between the Government of Nepal and UNDP, signed on 23 February 1984. All references in the SBAA to "Executing Agency" shall be deemed to refer to "Implementing Partner."

This project will be implemented by ***Alternative Energy Promotion Centre*** ("Implementing Partner") in accordance with its financial regulations, rules, practices and procedures only to the extent that they do not contravene the principles of the Financial Regulations and Rules of UNDP. Where the financial governance of an Implementing Partner does not provide the required guidance to ensure best value for money, fairness, integrity, transparency, and effective international competition, the financial governance of UNDP shall apply.

The designations employed and the presentation of material on this map do not imply the expression of any opinion whatsoever on the part of the Secretariat of the United Nations or UNDP concerning the legal status of any country, territory, city or area or its authorities, or concerning the delimitation of its frontiers or boundaries.

X. RISK MANAGEMENT

Government Entity (NIM)

1. Consistent with the Article III of the SBAA, the responsibility for the safety and security of the Implementing Partner and its personnel and property, and of UNDP's property in the Implementing Partner's custody, rests with the Implementing Partner. To this end, the Implementing Partner shall:
    a. put in place an appropriate security plan and maintain the security plan, taking into account the security situation in the country where the project is being carried;

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b. assume all risks and liabilities related to the Implementing Partner's security, and the full implementation of the security plan.

2. UNDP reserves the right to verify whether such a plan is in place, and to suggest modifications to the plan when necessary. Failure to maintain and implement an appropriate security plan as required hereunder shall be deemed a breach of the Implementing Partner's obligations under this Project Document.

3. The Implementing Partner agrees to undertake all reasonable efforts to ensure that no UNDP funds received pursuant to the Project Document are used to provide support to individuals or entities associated with terrorism, that the recipients of any amounts provided by UNDP hereunder do not appear on the United Nations Security Council Consolidated Sanctions List, and that no UNDP funds received pursuant to the Project Document are used for money laundering activities. The United Nations Security Council Consolidated Sanctions List can be accessed via https://www.un.org/securitycouncil/content/un-sc-consolidated-list.

4. The Implementing Partner acknowledges and agrees that UNDP will not tolerate sexual harassment and sexual exploitation and abuse of anyone by the Implementing Partner, and each of its responsible parties, their respective sub-recipients and other entities involved in Project implementation, either as contractors or subcontractors and their personnel, and any individuals performing services for them under the Project Document.

a) In the implementation of the activities under this Project Document, the Implementing Partner, and each of its sub-parties referred to above, shall comply with the standards of conduct set forth in the Secretary General's Bulletin ST/SGB/2003/13 of 9 October 2003, concerning "Special measures for protection from sexual exploitation and sexual abuse" ("SEA").

b) Moreover, and without limitation to the application of other regulations, rules, policies and procedures bearing upon the performance of the activities under this Project Document, in the implementation of activities, the Implementing Partner, and each of its sub-parties referred to above, shall not engage in any form of sexual harassment ("SH"). SH is defined as any unwelcome conduct of a sexual nature that might reasonably be expected or be perceived to cause offense or humiliation, when such conduct interferes with work, is made a condition of employment or creates an intimidating, hostile or offensive work environment. SH may occur in the workplace or in connection with work. While typically involving a pattern of conduct, SH may take the form of a single incident. In assessing the reasonableness of expectations or perceptions, the perspective of the person who is the target of the conduct shall be considered.

5. a) In the performance of the activities under this Project Document, the Implementing Partner shall (with respect to its own activities), and shall require from its sub-parties referred to in paragraph 4 (with respect to their activities) that they, have minimum standards and procedures in place, or a plan to develop and/or improve such standards and procedures in order to be able to take effective preventive and investigative action. These should include: policies on sexual harassment and sexual exploitation and abuse; policies on whistleblowing/protection against retaliation; and complaints, disciplinary and investigative mechanisms. In line with this, the Implementing Partner will and will require that such sub-parties will take all appropriate measures to:

i) Prevent its employees, agents or any other persons engaged to perform any services under this Project Document, from engaging in SH or SEA;

ii) Offer employees and associated personnel training on prevention and response to SH and SEA, where the Implementing Partner and its sub-parties referred to in paragraph 4 have not put in place its own training regarding the prevention of SH and SEA, the Implementing Partner and its sub-parties may use the training material available at UNDP;

iii) Report and monitor allegations of SH and SEA of which the Implementing Partner and its sub-parties referred to in paragraph 4 have been informed or have otherwise become aware, and status thereof;

iv) Refer victims/survivors of SH and SEA to safe and confidential victim assistance; and

v) Promptly and confidentially record and investigate any allegations credible enough to warrant an investigation of SH or SEA. The Implementing Partner shall advise UNDP of any such allegations received and investigations being conducted by itself or any of its sub-parties referred to in paragraph 4 with

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respect to their activities under the Project Document, and shall keep UNDP informed during the investigation by it or any of such sub-parties, to the extent that such notification (i) does not jeopardize the conduct of the investigation, including but not limited to the safety or security of persons, and/or (ii) is not in contravention of any laws applicable to it. Following the investigation, the Implementing Partner shall advise UNDP of any actions taken by it or any of the other entities further to the investigation.

b) The Implementing Partner shall establish that it has complied with the foregoing, to the satisfaction of UNDP, when requested by UNDP or any party acting on its behalf to provide such confirmation. Failure of the Implementing Partner, and each of its sub-parties referred to in paragraph 4, to comply of the foregoing, as determined by UNDP, shall be considered grounds for suspension or termination of the Project.

6. Social and environmental sustainability will be enhanced through application of the UNDP Social and Environmental Standards (http://www.undp.org/ses) and related Accountability Mechanism (http://www.undp.org/secu-srm).

7. The Implementing Partner shall: (a) conduct project and programme-related activities in a manner consistent with the UNDP Social and Environmental Standards, (b) implement any management or mitigation plan prepared for the project or programme to comply with such standards, and (c) engage in a constructive and timely manner to address any concerns and complaints raised through the Accountability Mechanism. UNDP will seek to ensure that communities and other project stakeholders are informed of and have access to the Accountability Mechanism.

8. All signatories to the Project Document shall cooperate in good faith with any exercise to evaluate any programme or project-related commitments or compliance with the UNDP Social and Environmental Standards. This includes providing access to project sites, relevant personnel, information, and documentation.

9. The Implementing Partner will take appropriate steps to prevent misuse of funds, fraud or corruption, by its officials, consultants, responsible parties, subcontractors and sub-recipients in implementing the project or using UNDP funds.

10. In the implementation of the activities under this Project Document, UNDP places reasonable reliance upon the Implementing Partner for it to apply its laws, regulations and processes, and applicable international laws regarding anti money laundering and countering the financing of terrorism, to ensure consistency with the principles of then in force the UNDP Anti-Money Laundering and Countering the Financing of Terrorism Policy.

11. The Implementing Partner will ensure that its financial management, anti-corruption, anti-fraud and anti money laundering and countering the financing of terrorism policies are in place and enforced for all funding received from or through UNDP.

12. The requirements of the following documents, then in force at the time of signature of the Project Document, apply to the Implementing Partner: (a) UNDP Policy on Fraud and other Corrupt Practices and (b) UNDP Office of Audit and Investigations Investigation Guidelines. The Implementing Partner agrees to the requirements of the above documents, which are an integral part of this Project Document and are available online at www.undp.org.

13. In the event that an investigation is required, UNDP has the obligation to conduct investigations relating to any aspect of UNDP projects and programmes in accordance with UNDP's regulations, rules, policies and procedures. The Implementing Partner shall provide its full cooperation, including making available personnel, relevant documentation, and granting access to the Implementing Partner's (and its consultants', responsible parties', subcontractors' and sub-recipients') premises, for such purposes at reasonable times and on reasonable conditions as may be required for the purpose of an investigation. Should there be a limitation in meeting this obligation, UNDP shall consult with the Implementing Partner to find a solution.

14. The signatories to this Project Document will promptly inform one another in case of any incidence of inappropriate use of funds, credible allegation of fraud or corruption or other financial irregularities with due confidentiality.

15. Where the Implementing Partner becomes aware that a UNDP project or activity, in whole or in part, is the focus of investigation for alleged fraud/corruption, the Implementing Partner will inform the UNDP Resident Representative/Head of Office, who will promptly inform UNDP's Office of Audit and Investigations (OAI). The

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Implementing Partner shall provide regular updates to the head of UNDP in the country and OAI of the status of, and actions relating to, such investigation.

16. The Implementing Partner agrees that, where applicable, donors to UNDP (including the Government) whose funding is the source, in whole or in part, of the funds for the activities which are the subject of this Project Document, may seek recourse to the Implementing Partner for the recovery of any funds determined by UNDP to have been used inappropriately, including through fraud, corruption or other financial irregularity, or otherwise paid other than in accordance with the terms and conditions of the Project Document.

Where such funds have not been refunded to UNDP, the Implementing Partner agrees that donors to UNDP (including the Government) whose funding is the source, in whole or in part, of the funds for the activities under this Project Document, may seek recourse to the Implementing Partner for the recovery of any funds determined by UNDP to have been used inappropriately, including through fraud, corruption or other financial irregularity, or otherwise paid other than in accordance with the terms and conditions of the Project Document.

Note: The term "Project Document" as used in this clause shall be deemed to include any relevant subsidiary agreement further to the Project Document, including those with responsible parties, subcontractors, and sub-recipients.

17. Each contract issued by the Implementing Partner in connection with this Project Document shall include a provision representing that no fees, gratuities, rebates, gifts, commissions or other payments, other than those shown in the proposal, have been given, received, or promised in connection with the selection process or in contract execution, and that the recipient of funds from the Implementing Partner shall cooperate with any and all investigations and post-payment audits.

18. Should UNDP refer to the relevant national authorities for appropriate legal action any alleged wrongdoing relating to the project, the Government will ensure that the relevant national authorities shall actively investigate the same and take appropriate legal action against all individuals found to have participated in the wrongdoing, recover and return any recovered funds to UNDP.

19. The Implementing Partner shall ensure that all of its obligations set forth under this section entitled "Risk Management" are passed on to each responsible party, subcontractor and sub-recipient and that all the clauses under this section entitled "Risk Management Standard Clauses" are included, mutatis mutandis, in all sub-contracts or sub-agreements entered into further to this Project Document.

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XI. ANNEXES

ANNEX 1: PROJECT QUALITY ASSURANCE REPORT

ANNEX 2: SOCIAL AND ENVIRONMENTAL SCREENING

Social and Environmental Screening Template (2021 SESP Template, Version 1)

The completed template, which constitutes the Social and Environmental Screening Report, must be included as an annex to the Project Document at the design stage. Note: this template will be converted into an online tool. The online version will guide users through the process and will embed relevant guidance.

Project Information

Project Information	
1. Project Title	**Renewable Energy for Rural Livelihood Programme (RERL)**
2. Project Number (i.e. Atlas project ID, PIMS+)	00076958
3. Location (Global/Region/Country)	Nepal
4. Project stage (Design or Implementation)	Implementation
5. Date	13 March 2022

Part A. Integrating Programming Principles to Strengthen Social and Environmental Sustainability

QUESTION 1: How Does the Project Integrate the Programming Principles in Order to Strengthen Social and Environmental Sustainability?
Briefly describe in the space below how the project mainstreams the human rights-based approach
The AEPC/RERL project aims to increase access to electricity and enhance renewable energy based rural enterprises development in selected rural communities of Nepal, especially in off-grid areas of the country. This is inline with the citizen's right to the basic need in accordance with the Constitution of Nepal. The Project envisions ensures reliable supply of energy in an affordable and easy manner, and make proper use of energy, for the fulfilment of the basic needs of citizens, by generating and developing renewable energy. The Project thus envisions participation of women, vulnerable, disadvantaged and socially excluded groups into every project cycle of the subprojects and promote equal opportunities to women and men at the time of construction of RE systems. The Environment Assessment and Review Framework and Indigenous Peoples Planning Framework implemented by the UNDP will ensure compliance with environmental and social safeguard principles for SASEC.
Briefly describe in the space below how the project is likely to improve gender equality and women's empowerment
AEPC/RERL designs every energy sub-project in a way that ensures access to electricity for women from which they can have socio-economic benefit. The subproject has a mandatory requirement to have at least 30% women in decision-making bodies and capacity development activities. The GESI Action Plan will be prepared and implemented in all stages of the project cycle and UNDP will facilitate direct engagement of women and members of vulnerable, disadvantaged groups in productive end use activities.
Briefly describe in the space below how the project mainstreams sustainability and resilience
The Project ensures the local rural/municipality and local stakeholders including such as potential beneficiaries, local entrepreneurs, cooperative/company etc. to ascertain demand and resources and ownership of such systems. The detailed engineering design study of the subproject will be carried out only after receiving firm commitment of upfront equity by project developer. Post installation support will be provided to include both institutional strengthening for smooth operation of the plants and promotion of productive electricity uses. A Special Purpose Vehicle (SPV) will be setup that will develop, own and manage subprojects which will be registered as a cooperative or a company (private/public). Support will be extended

Risk 2: activities within or adjacent to critical habitats and/or environmentally sensitive areas, including (but not limited to) legally protected areas (e.g. nature reserve, national park), areas proposed for protection, or recognized as such by authoritative sources and/or indigenous peoples or local communities	I = 3 L = 3	**M**	Some of the subproject under AEPC/RERL are in buffer zone of the national park and one is adjacent to the hunting reserve.	- Initial Environmental Examination has been conducted for such subprojects and required GoN approval has been obtained.
Risk 3: outputs and outcomes sensitive or vulnerable to potential impacts of climate change or disasters	I = 3 L = 2	**L**	Disasters might affect the infrastructure of the subprojects and disrupt the access to energy	- Integration of protective measures as well as Early warning system, Disaster resilient infrastructure in subproject design
Risk 4: construction and/or infrastructure development	I = 2 L = 1	**L**	Unplanned construction activities and inadequate safety measures at the construction site might pose risk to the community	- Preparation of Environmental and Social Management Plan, including construction plan as well as community health and safety plan
Risk 5: influx of project workers to project areas	I = 1 L =3	**L**	Usually the subprojects recruit unskilled and semiskilled human resources (workers) for the local community. However, a few skilled workers shall be recruited from other areas.	- Coordination with the developer and local communities will be facilitated.
Risk 6: areas where indigenous peoples are present	I = 1 L = 3	**L**	Most of the subprojects being in off grid remote area, is resided by indigenous peoples.	- Indigenous Peoples Plan Framework of ADB/SASEC provides guidance on their enhancement
Risk 7: occupational health and safety risks due to physical, chemical, biological and psychosocial hazards (including violence and harassment) throughout the project life-cycle	I = 2 L =2	**L**	The nature of work such as excavation, work in confined space and in height poses physical hazard. In addition, working away from home for migrant worker might be stressful.	- Formulation and implementation of Occupational Health and Safety Plan, Labor Management Plan
Risk 8: the generation of waste (both hazardous and non-hazardous)	I = 2 L =2	**L**	Spoil generated from excavation, packaging waste, waste from Labor camp etc	- Mitigation measures are suggested in the subproject specific Environmental and Social Management Plan - Waste management plan will be developed
	QUESTION 4: What is the overall project risk categorization?			
	Low Risk	✓	The overall Project Risk is categorized as **Low Risk** in line with the subsequent mechanism have been setup and proposed by the Project. With the mandatory	

	6. *Indigenous Peoples*	✓	
	7. *Labour and Working Conditions*	✓	
	8. *Pollution Prevention and Resource Efficiency*	✓	

Final Sign Off

Final Screening at the design-stage is not complete until the following signatures are included

Signature	*Date*	*Description*
QA Assessor DocuSigned by: B801579101E84F7...	14-Mar-2022	UNDP staff member responsible for the project, typically a UNDP Programme Officer. Final signature confirms they have "checked" to ensure that the SESP is adequately conducted.
QA Approver DocuSigned by: 6DDBE95E4F9B4E9...	14-Mar-2022	UNDP senior manager, typically the UNDP Deputy Country Director (DCD), Country Director (CD), Deputy Resident Representative (DRR), or Resident Representative (RR). The QA Approver cannot also be the QA Assessor. Final signature confirms they have "cleared" the SESP prior to submittal to the PAC.
PAC Chair		UNDP chair of the PAC. In some cases PAC Chair may also be the QA Approver. Final signature confirms that the SESP was considered as part of the project appraisal and considered in recommendations of the PAC.

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Accountability		
Would the project potentially involve or lead to:		
P.13	exclusion of any potentially affected stakeholders, in particular marginalized groups and excluded individuals (including persons with disabilities), from fully participating in decisions that may affect them?	*No*
P.14	grievances or objections from potentially affected stakeholders?	*No*
P.15	risks of retaliation or reprisals against stakeholders who express concerns or grievances, or who seek to participate in or to obtain information on the project?	*No*
Project-Level Standards		
Standard 1: Biodiversity Conservation and Sustainable Natural Resource Management		
Would the project potentially involve or lead to:		
1.1	adverse impacts to habitats (e.g. modified, natural, and critical habitats) and/or ecosystems and ecosystem services? *For example, through habitat loss, conversion or degradation, fragmentation, hydrological changes*	*No*
1.2	activities within or adjacent to critical habitats and/or environmentally sensitive areas, including (but not limited to) legally protected areas (e.g. nature reserve, national park), areas proposed for protection, or recognized as such by authoritative sources and/or indigenous peoples or local communities?	*Yes*
1.3	changes to the use of lands and resources that may have adverse impacts on habitats, ecosystems, and/or livelihoods? (Note: if restrictions and/or limitations of access to lands would apply, refer to Standard 5)	*No*
1.4	risks to endangered species (e.g. reduction, encroachment on habitat)?	*No*
1.5	exacerbation of illegal wildlife trade?	*No*
1.6	introduction of invasive alien species?	*No*
1.7	adverse impacts on soils?	*No*
1.8	harvesting of natural forests, plantation development, or reforestation?	*No*
1.9	significant agricultural production?	*No*
1.10	animal husbandry or harvesting of fish populations or other aquatic species?	*No*
1.11	significant extraction, diversion or containment of surface or ground water? *For example, construction of dams, reservoirs, river basin developments, groundwater extraction*	*No*
1.12	handling or utilization of genetically modified organisms/living modified organisms?[2]	*No*
1.13	utilization of genetic resources? (e.g. collection and/or harvesting, commercial development)[3]	*No*
1.14	adverse transboundary or global environmental concerns?	*No*
Standard 2: Climate Change and Disaster Risks		
Would the project potentially involve or lead to:		
2.1	areas subject to hazards such as earthquakes, floods, landslides, severe winds, storm surges, tsunami or volcanic eruptions?	*No*
2.2	outputs and outcomes sensitive or vulnerable to potential impacts of climate change or disasters?	*Yes*

[2] See the Convention on Biological Diversity and its Cartagena Protocol on Biosafety.
[3] See the Convention on Biological Diversity and its Nagoya Protocol on access and benefit sharing from use of genetic resources.

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5.3	risk of forced evictions?[4]	*No*
5.4	impacts on or changes to land tenure arrangements and/or community based property rights/customary rights to land, territories and/or resources?	*No*
Standard 6: Indigenous Peoples		
Would the project potentially involve or lead to:		
6.1	areas where indigenous peoples are present (including project area of influence)?	*Yes*
6.2	activities located on lands and territories claimed by indigenous peoples?	*No*
6.3	impacts (positive or negative) to the human rights, lands, natural resources, territories, and traditional livelihoods of indigenous peoples (regardless of whether indigenous peoples possess the legal titles to such areas, whether the project is located within or outside of the lands and territories inhabited by the affected peoples, or whether the indigenous peoples are recognized as indigenous peoples by the country in question)? *If the answer to screening question 6.3 is "yes", then the potential risk impacts are considered significant and the project would be categorized as either Substantial Risk or High Risk*	*No*
6.4	the absence of culturally appropriate consultations carried out with the objective of achieving FPIC on matters that may affect the rights and interests, lands, resources, territories and traditional livelihoods of the indigenous peoples concerned?	*No*
6.5	the utilization and/or commercial development of natural resources on lands and territories claimed by indigenous peoples?	*No*
6.6	forced eviction or the whole or partial physical or economic displacement of indigenous peoples, including through access restrictions to lands, territories, and resources? *Consider, and where appropriate ensure, consistency with the answers under Standard 5 above*	*No*
6.7	adverse impacts on the development priorities of indigenous peoples as defined by them?	*No*
6.8	risks to the physical and cultural survival of indigenous peoples?	*No*
6.9	impacts on the Cultural Heritage of indigenous peoples, including through the commercialization or use of their traditional knowledge and practices? *Consider, and where appropriate ensure, consistency with the answers under Standard 4 above.*	*No*
Standard 7: Labour and Working Conditions		
Would the project potentially involve or lead to: (note: applies to project and contractor workers)		
7.1	working conditions that do not meet national labour laws and international commitments?	*No*
7.2	working conditions that may deny freedom of association and collective bargaining?	*No*
7.3	use of child labour?	*No*
7.4	use of forced labour?	*No*
7.5	discriminatory working conditions and/or lack of equal opportunity?	*No*
7.6	occupational health and safety risks due to physical, chemical, biological and psychosocial hazards (including violence and harassment) throughout the project life-cycle?	*Yes*
Standard 8: Pollution Prevention and Resource Efficiency		
Would the project potentially involve or lead to:		

[4] Forced eviction is defined here as the permanent or temporary removal against their will of individuals, families or communities from the homes and/or land which they occupy, without the provision of, and access to, appropriate forms of legal or other protection. Forced evictions constitute gross violations of a range of internationally recognized human rights.

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ANNEX 3: RISK ANALYSIS

Offline Project Risk Register for Project Document

Project Title: Renewable Energy for Rural Livelihood **Award ID/Project Number: 00076958/00117173** **Date: 25 April 2022**

#	Description	Risk Category	Impact & Likelihood = Risk Level	Risk Treatment / Management Measures	Risk Owner
1	Construction of MHPs not completed within the original project period of December 2021 due to various reasons incl. delay in timely initiation, community conflicts, delay in financial closure	Operational	The delay in completing construction activities in 4 mini hydro projects (Khatyad, Chukeni, Saniveri and Ankhe) will have adverse effects in meeting the project targets by December 2021 P = 5; I = 5 Risk Level = High	Ministry of Finance has submitted formal request for time extension of the SASEC project up to the end of 2023 along with additional resources requirement	AEPC, RERL
2	Lack of financial capacity of contractors to complete mini hydro projects on time	Operational	The contractors selected through the PPA/PPR procurement process by AEPC for construction of mini hydro projects have difficulties in maintaining the cash flow needed to complete the construction activities on time. This will negatively impact the project delivery in scheduled time while also having implications on cost overrun. P = 4; I = 5 Risk Level = High	The project is facilitating provision of credit facility from Central Renewable Energy Fund to maintain necessary cash flow to needy contractor. Similarly, the project is building capacity of contractors. Vendors and site engineers on different aspects of project construction management.	AEPC, RERL
3	Arrival of national grid at the project sites that were off grid during the project initiation and construction	Regulatory	Due to rapid expansion of the national grid, the mini hydro projects in off-grid areas during development have the risk of grid encroachment. This results in risks on future sustainability of the isolated RE projects promoted. P = 4; I = 2 Risk Level = Moderate	The project is supporting developers for grid connection of mini grids. Simrutu Khola & Ankhe Khola MHPs have signed Net Metering Agreement with NEA & documents for net metering of Giri Khola & Middle Phawa Khola have been submitted to NEA. Meanwhile, AEPC and NEA have decided to develop Saniveri MHP in very close coordination where AEPC will be responsible for the generation component while NEA will build the distribution network with its own resources. Similarly, AEPC and NEA are preparing an Interconnection Master Plan for Mini Grids to facilitate in the policy direction for grid interconnection.	AEPC, NEA, RERL

4	Due to the scale of the projects that are being developed, there is distinct lack of capacity of community to manage the community-based projects	Organizational	The lack of capacity to run and manage the developed mini hydro projects, there is likelihood that regular electricity supply and services can be disrupted in the future. This results in weak revenue stream and risks to lack of proper repair and maintenance of the projects P = 4; I = 5 Risk Level = High	The project is building capacity of developers for establishment of accountable governance and management system with incentive-based rules and regulations. RERL has initiated establishment of a mechanism for technical backstopping by experienced professionals as a key aspect of post installation support through the project.	RERL
5	Natural Calamities (earthquake, flood, landslide, etc.)	Environmental	The risk, beyond the control of the project, hampers timely project delivery. This will also have implications on the need for reconstruction and rehabilitation of the project that will incur additional costs thus increasing the project costs significantly. P = 3; I = 5 Risk Level = Substantial	The project will ensure that mitigation measures identified in environmental assessment will be implemented to minimize damages due to natural calamities if and when that occurs.	AEPC, RERL
6	Lockdown due to COVID-19 Pandemic	Health and Safety	The difficulties in mobilizing human resources and materials to the sites during the pandemic delays timely project delivery while also increasing the project costs. P = 5; I = 4 Risk Level = High	To minimize this risk, Standard Operating Procedure has been prepared and shared with field staff. Orientation has also been provided to the field staff, Site Engineers, Mobilizers, construction workers & community in general The project has supported provision of health and safety measures including masks and sanitizer in all project sites. The project is facilitating with contractors to establish separate/isolated living quarters for construction personnel for uninterrupted works	RERL

ANNEX 4: CAPACITY ASSESSMENT

HACT Micro Assessment was conducted in 2018 and is available at Micro Assessment Report_AEPC_2018.pdf

ANNEX 5: PROJECT BOARD TERMS OF REFERENCE AND TORS OF KEY MANAGEMENT POSITIONS

UNDP Standard Terms of Reference (ToR) for the Project Board of RERL

I. Background

All UNDP projects must be governed by a multi-stakeholder board or committee established to review performance based on established monitoring and evaluation metrics and high-level implementation issues to ensure quality delivery of results. For the purpose of this ToR and to ensure standardization, henceforth, as regards project documentation, such a body shall only be referred to by 'Project Board'. The Project Board is the most senior, dedicated oversight body for a UNDP 'Development Project', which is defined in the PPM as an instrument where UNDP "Delivers outputs where UNDP has accountability for design, oversight and quality assurance of the entire project."

II. Duties and Responsibilities

The two prominent (mandatory) roles of the Project Board are as follows:

1) **High-level oversight of the project** (as explained in the "Provide Oversight" section of the PPM). This is the primary function of the Project Board. The Project Board reviews evidence of project performance based on monitoring, evaluation and reporting, including progress reports, monitoring missions' reports, evaluations, risk logs, quality assessments, and the combined delivery report. The Project Board is the main body responsible for taking corrective actions as needed to ensure the project achieves the desired results. And its function includes oversight of annual (and as-needed) assessments of any major risks to the programme or project, and related decisions/agreements on any management actions or remedial measures to address them effectively.

   The Project Board also carries the role of quality assurance of the project taking decisions informed by, among other inputs, the project quality assessment. In this role the Board is supported by the quality assurer, whose function is to assess the quality of the project against the corporate standard criteria. This function is performed by a UNDP programme or monitoring and evaluation officer to maintain independence from the project manager regardless of the project 's implementation modality.

   The Project Board reviews updates to the project risk log.

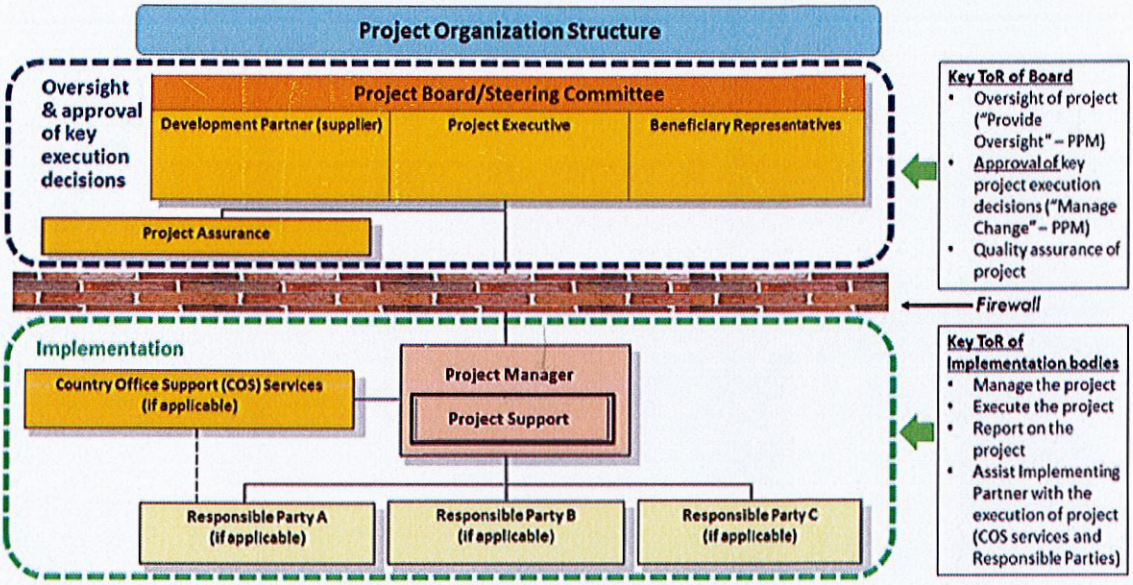
2) **Approval of key project execution decisions** (as explained in the "Manage Change" section of the PPM). The Project Board has an equally important, secondary role in approving certain adjustments above provided tolerance levels, including substantive programmatic revisions (major/minor amendments), budget revisions, requests for suspension or extension and other major changes (subject to additional funding partner/donor requirements).

The Project Board is responsible for making management decisions by consensus when required, including the approval of project plans and revisions, and the project manager's tolerances. The Project Board approves annual work plans and reviews updates to the project risk log.

Within the overall governance and management arrangements of the project, the role of the Project Board as regards these two key functions ('High-level oversight of the project' and 'Approval of key project execution decisions') is distinct from the roles of entities involved in the implementation of the project, namely the implementing partner (IP), responsibilities parties (if applicable), service providers and project staff.

The diagram below outlines the main entities involved (and their respective responsibilities) in the 'oversight/approval of key execution decisions' layer and the 'implementation' layer of the project structure.

Diagram 1 – Standard Figure of Project Organization Structure vis-à-vis oversight & approval and implementation roles



In cases where UNDP or a national government entity are concurrently playing roles and represented in both layers of the project organization structure, the entity must seek to separate its project oversight and implementation duties and describe in the relevant project document a: 1) satisfactory internal institutional arrangement for the separation of oversight and implementation functions in different departments of said entity and; 2) clear lines of responsibility, reporting and accountability within the entity between their oversight and implementation functions.

In order to ensure UNDP's ultimate accountability, the Project Board decisions should be made in accordance with the Quality Standards for Programming that shall ensure management for development results, best value money, fairness, integrity, transparency and effective national and international competition. An effective Project Board needs credible data, evidence, quality assurance and reporting to aid decision making (see next section on supporting functions to the Board). The Project Board also needs to be accountable to protect against conflicts of interest and fraud.

Specific responsibilities of the Project Board include the following:

- Provide overall guidance and direction to the project, ensuring it remains within any specified constraints, and promote gender equality and social inclusion (LNOB) in the project implementation;
- Review project performance based on monitoring, evaluation and reporting, including standard quality assurance checks, progress reports, risk logs, spot checks/audit reports and the combined delivery report;
- Address any high-level project issues as raised by the project manager and project assurance;
- Provide guidance on emerging and/or pressing project risks and agree on possible mitigation and management actions to address specific risks (including ensuring compliance with UNDP's Social and Environmental Standards, Fraud/corruption, Sexual Exploitation and Abuse and Sexual Harassment);
- Agree or decide on project manager's tolerances as required, within the parameters set by UNDP (Manage Change in the PPM) and the donor, and provide direction and decisions for exceptional situations when the project manager's tolerances are exceeded;
- Advise on major and minor amendments to the project within the parameters set by UNDP and the donor;
- Agree or decide on a project suspension or cancellation, if required; (note that for GEF and GCF projects it is UNDP that decides to suspend or cancel and project and the Project Board is informed/consulted only).
- Provide high-level direction and recommendations to the project management unit to ensure that the agreed deliverables are produced satisfactorily according to plans.
- Receive and address project level grievance, including overseeing whatever specific compliance and stakeholder response (or grievance) mechanisms have been put in place so that individuals and communities

potentially affected by the project have access to effective mechanisms and procedures for raising concerns about the social and environmental performance of the project[6].

- Engage in the low value grant selection process where there is no Grant Selection Committee, as guided by the Low Value Grants – UNDP Operational Guide.

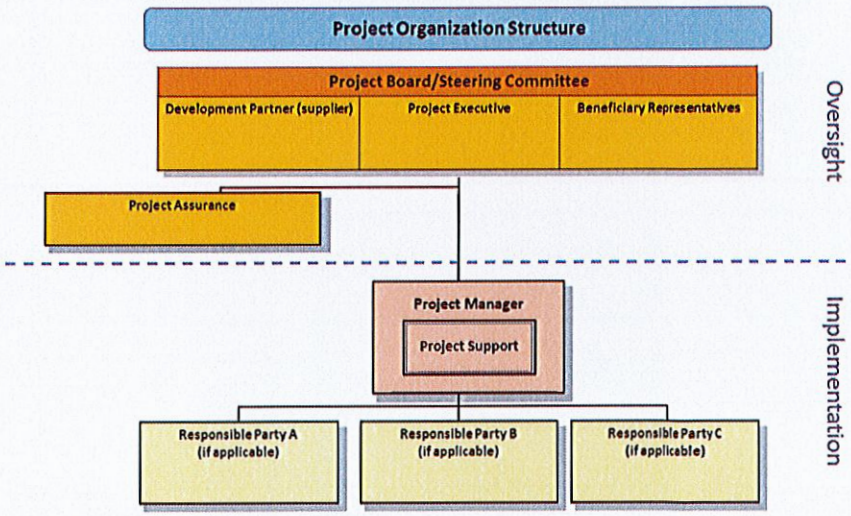
Additional responsibilities of the Project Board can include, but are not limited to, the following:

- Ensure coordination between the various donors and government-funded projects and programmes;
- Report to relevant inter-ministerial bodies or higher-level oversight bodies;
- Ensure coordination with multiple government agencies and their participation in project activities;
- Track and monitor co-financing for this project;
- Appraise the annual project implementation report, including the quality assessment rating report;
- Ensure commitment of human resources to support project implementation, arbitrating any issues within the project;
- Act as an informal consultation mechanism for stakeholders;
- Approve the Project Inception Report, Mid-term Review and Terminal Evaluation reports and corresponding management responses;
- Review the final project report package during an end-of-project review meeting to discuss lessons learned and opportunities for scaling up;
- Providing guidance or reporting protocols to technical committees or sub-bodies reporting to the Board (if applicable);

III. Composition of the Project Board

As noted in the diagram below, every Project Board in a UNDP project has three categories of formal members (e.g. voting members). The role of every formal Project Board member must correspond to one of these three roles and be identified accordingly in the project documentation.

Diagram 2 – Standard Figure for a Project Organization Structure



The three categories of Project Board members are the following:

1) **Project Director/Executive(s):** This is an individual(s) who represents ownership of the project and chairs (or co-chairs) the Project Board. The executive usually is the most senior national counterpart for nationally

[6] The responsibilities of the board in this regard should follow UNDP's Social and Environmental Standards (SES) as codified in the PPM. It should be noted that while a project board can play a role in addressing or assisting with the compliance and stakeholder response (or grievance) mechanisms put in in place for a given project (as part of their quality assurance and oversight function), this will be in addition to and does not substitute for UNDP's core responsibility to ensure compliance with the SES throughout the project management cycle as part of UNDP's Programming Quality Assurance system.

implemented projects (typically from the same entity as the Implementing Partner), and it must be UNDP for projects that are direct implementation (DIM). In exceptional cases, two individuals from different entities can co-share this role and/or co-chair the Project Board. If the project executive co-chairs the Project Board with a representative of another category, it typically does so with a development partner representative. The Project Executive is the Executive Director of Alternative Energy Promotion Centre.

2) **Beneficiary Representative(s):** This is an individual(s) representing the interests of those groups of stakeholders who will ultimately benefit from the project. Their primary function within the Board is to ensure the realization of project results from the perspective of project beneficiaries. Often representatives from civil society, industry associations, community groups or other government entities benefiting from the project can fulfil this role. If the project has a specific geographic focus, often representatives from the government entities in the targeted area/region will play this role. There can be multiple beneficiary representatives in a Project Board. The Beneficiary Representative(s) is the Ministry of Energy Water Resources and Irrigation, Ministry of Finance, Local Governments and the private sector.

3) **Development Partner(s):** Individuals representing the interests of the parties concerned that provide funding, strategic guidance and/or technical expertise to the project[7]. There can be multiple development partners representatives in a Project Board. The Development Partner(s) representative(s) is the Senior Project Officer (Energy) of Asian Development Bank and the Deputy Resident Representative of UNDP Nepal. Other funding partners can also jointly hold this role. As noted below, responsible parties cannot play this role)

A UNDP representative must always be represented in the Project Board in either the project executive or development partner role. The level of representation by UNDP on the Board is determined on a case-by-case basis but typically, for national projects, the board member is either the Deputy Resident Representative or Assistant Resident Representative.]

Where applicable, representatives from responsible parties to the project cannot sit on the Project Board as a formal voting member; they can (if requested) attend board meetings as observers. Since the chief responsibility of the Board is to provide high-level oversight of project implementation, to avoid any conflicts of interest, it is not appropriate for representatives of third party entities engaged by the project to provide services – whether responsible parties or contractors/service providers – to concurrently sit on the Board. Representatives of responsible parties can attend board meetings (as observers) but can have no official role in board decision-making. The same principle applies to the project manager who in attending and presenting at board meetings, does so in a non-voting capacity.

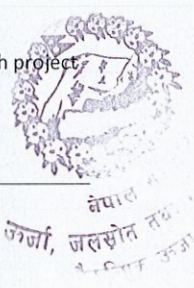
In cases where the inputs and guidance of responsible parties or other entities formally engaged in providing goods or services to the project are needed by the Board on a recurring basis, it is recommended to establish appropriate advisory or technical committees or working groups composed of those entities that can formally report to the Board, while ensuring the impartiality and integrity of board decisions happening independent of those bodies (see Section V of the ToR).

IV. Standard Project Board Protocols

The Project Board must meet one time annually at a minimum. It is recommended that the timing of board meetings be agreed upon in advance and corresponds to key project reporting or work planning milestones. This Project Board will meet once every quarter according to this provisional schedule.

Project Board members cannot receive remuneration from project funds for their participation in the Board. However, it is allowable for board members to be reimbursed from project funds for certain reasonable, qualified expenses related to travel or lodging to attend board meetings. Such protocols are outlined in this ToR and the benefits are applicable to all eligible board members.

[7] With the exception of responsible parties or any firms/entities engaged by the project to provide technical expertise with project funds



All Project Boards must have rules for quorum and documentation/minuting of board decisions. All board decisions and minutes should be kept by the project management unit and UNDP. Guidelines on decisions taken in between board meetings or virtually should be clearly elaborated in the Terms of Reference (ToR) of the Project Board.

Unless otherwise specified, Project Board decisions are made by unanimous consensus. If a consensus cannot be reached within the Board, the final decision shall rest with the UNDP representative on the Project Board or a UNDP staff member with delegated authority as the programme manager.[8]

It is required that as per internationally recognized professional standards and principles of sound governance, conflicts of interest affecting board members in performing their duties must be formally disclosed if not avoidable. Where a board member has a specific personal conflict of interest with a given matter before the board, he/she must recuse oneself from their participation in a decision. No board member can vote or deliberate on a question in which he/she has a direct personal or pecuniary interest not common to other members of the board.

All board members should be presented with a ToR for the Project Board, which will include the responsibilities already outlined and indicate agreed board practices and logistics.

Note: Designated Board members must sign or otherwise indicate written acceptance of the final ToR (via formal signature of the ToR, an exchange of letters or e-mail acknowledgment) for the Project Board as a precondition for serving on the Project Board. Please see Annex A for possible options to codify this acceptance (which preferably would be done at the appraise and approve stage but which could also be done early in implementation). The formal written acceptance of the ToR by all board members will be documented and kept by UNDP.

V. Standard Outputs of Project Board Meetings

In its oversight function, the Project Board will (at a minimum) review and assess the following project-related evidence at each meeting:

- Assessment of project progress to date against project output indicators (as documented in the project document results framework)
- Approval/review of annual work plans
- Assessment of the relevant Monitoring & Evaluation mechanisms, including all evaluations[9]
- Review and assessment of the Project Risk Log (with updating/amendments as needed)
- Assessment of project spending, based on a review of the combined delivery report
- Review of required resources versus available funding (if applicable) and steps taken to reduce funding gap identified at the project design stage

This will be in addition to the review and approval of any required project execution decisions.

The output of every Project Board should be a written record (minutes) that captures the agenda and issues discussed and the agreed upon action items and decisions (if applicable). Each report should clearly document the members attending the meeting (as well as all participants in the meeting) and the modality used to agree on a certain action or decision (whether formal voting or no-objection or other mechanism). All records of board meetings should be documented and kept by UNDP in their quality assurance function (see next section).

VI. Support Functions to the Project Board

There are two main entities/functions outside the Project Board structure whose role is to report to the Project Board and support board members in effectively fulfilling their roles: project assurance and project management.

The diagram below explains the primary role of these two entities in the context of their support to Board operations. A description of these two entities follows.

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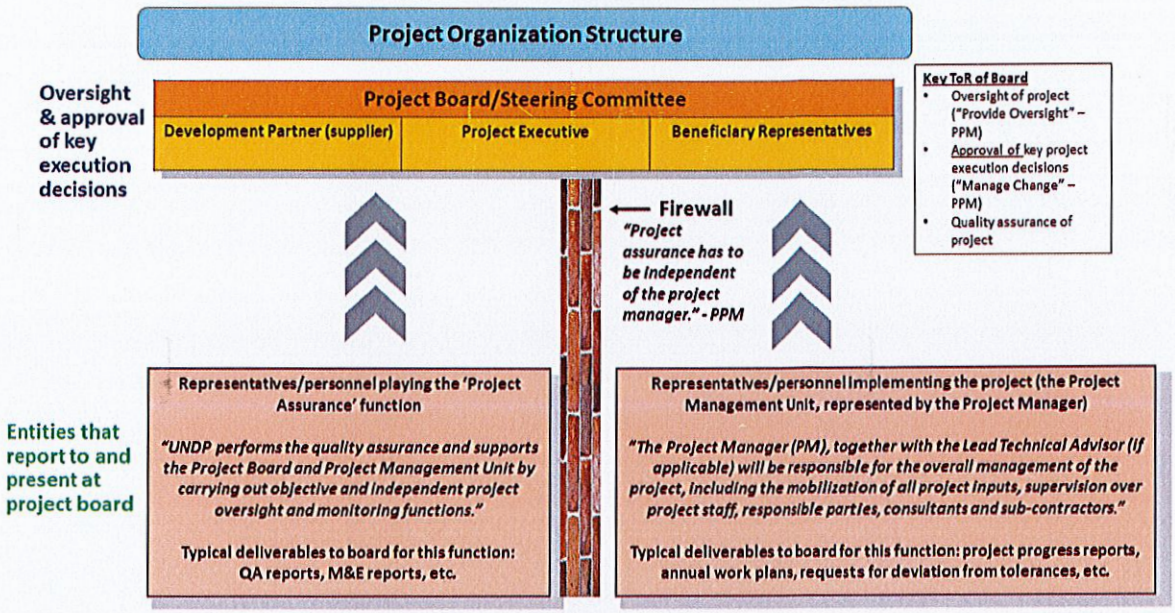
[8] UNDP has this special right since the ultimate legal and fiduciary accountability for a UNDP project, irrespective of modality, rests with UNDP and UNDP must (in line with its obligations to donors and to the Executive Board) be able to ensure that no action is taken by any body in a UNDP project that contravenes UNDP rules and regulations.

[9] Including audit reports and spot checks.

Diagram 3 – Standard Figure of Project Organization Structure – Board Support Entities



Project Assurance: Project assurance is the responsibility of each Project Board member; however, UNDP has a distinct assurance role for all UNDP projects in carrying out objective and independent project oversight and monitoring functions. UNDP performs quality assurance and supports the Project Board (and Project Management Unit) by carrying out objective and independent project oversight and monitoring functions, including applying UNDP's social and environmental management system to ensure the SES are applied through the project cycle. The Project Board cannot delegate any of its quality assurance responsibilities to the project manager. Project assurance is totally independent of project execution. [In the case where UNDP is also supporting project implementation, protocols should be put in place ensuring proper segregation between UNDP's role in the Board and in the function of project assurance, and any concurrent role in implementation].

A designated representative of UNDP playing the project assurance role is expected to attend all Project Board meetings and support board processes as a non-voting representative. It should be noted that while in certain cases UNDP's project assurance role across the project may encompass activities happening at several levels (e.g. global, regional), at least one UNDP representative playing that function must, as part of their duties, specifically attend board meetings and provide board members with the required documentation required to perform their duties.

The UNDP representative playing the main project assurance function is typically CO Portfolio Manager.

Project Support, this function is often covered by the Project Management Unit: The Project Manager (PM) (also called project coordinator) is the senior most representative of the Project Management Unit (PMU) and is responsible for the overall management of the project on behalf of the Implementing Partner, including the mobilization of all project inputs, supervision over project staff, responsible parties, consultants and sub-contractors. The project manager typically presents key deliverables and documents to the Board for review and approval, including progress reports, annual work plans, adjustments to tolerance levels and risk logs.

A designated representative of the PMU is expected to attend all board meetings and present the required progress reports and other documentation needed to support board processes as a non-voting representative.

The primary PMU representative attending board meetings is the National Programme Manager of the project.

Annex A – Acknowledgement of this ToR by each designated official Project Board member

Note: The form/evidence for this acknowledgement must be included as an annex or codified in other ways (e.g. note to file or minutes of a board meeting)

The formal acknowledgement of the ToR by board members can be done via various procedures, including the following options:

1) Having copies of the ToR be signed by each appointed board member at the Appraise and Approve stage (LPAC) and then having all signed copies attached as an annex to the Project Document
2) Having the final ToR be signed by each appointed board member at the first project board meeting after Project Document signing with the approvals recorded in the minutes of the meeting

In both cases, the signature or acknowledgement should include the name, title and category of board representation for the person signing.

Terms of Reference (ToR)

National Programme Manager/Senior National Advisor

The National Programme Manager (NPM), who also will be the Senior National Advisor (SNA) of the project, will be responsible for the overall management of the project implementation which includes personnel, subcontracts, training, equipment, administrative support and financial reporting. The NPM will carry out project activities under the guidance and supervision of AEPC. More specific responsibilities will be:

- Prepare periodic work plans integrated with the corresponding plans for AEPC.
- Prepare Annual Work Plan with detail budget allocation based on the Project Allocation of the project and submit it for approval.
- Ensure timely mobilization and utilization of project personnel, subcontracts, training and equipment inputs, whether these are procured by the project itself or by other agencies.
- Exercise overall technical, financial and administrative oversight of the project, including supervision of personnel assigned to the project.
- Implement activities related to the human resources development, community mobilization, private sector development, organizational strengthening and GESI.
- Maintain close cooperation with stakeholders in implementing the project activities.
- Prepare the Individual Performance Plans and Performance Appraisal of all staff.
- Carry out regular follow-up and monitoring to the communities and facilitate joint monitoring visits from AEPC.
- Ensure timely preparation and submission of required reports, including technical, financial, study reports.
- Support and promote GESI in project activities as well as among the project staff.

Furthermore, the SNA will carry out the following tasks:

- Support AEPC in its RE promotion activities.
- Ensure cross-component coordination including Central Renewable Energy Fund (CREF) related to activities of the project.
- Ensure close coordination with government and other stakeholder through AEPC to achieve better synergy.
- Support AEPC for policy formulation and advocacy for development of mini hydro and solar and solar wind hybrid mini grid system development.
- Participate and encourage participation of project staff in different forums that may be organized by various agencies related to the mini hydro and solar mini grid projects and other related subjects.
- Support AEPC to take lead role in networking meetings amongst key stakeholders of the RE projects and other government and donor agencies.
- Document and disseminate lessons learned through studies, audio visual production and electronic means.
- Provide additional support as requested and required by the GoN and TA Service Provider to make the project a success.
- Participate in meetings, trainings, workshops and events.

Qualifications: The NPM/SNA should have a Master's degree in Engineering, Management, Energy or a related field. The NPM/SNA should preferably have 12 years of working experience in RE with preferably 5 years of successful TA managerial experience. The NPM/SNA should have relevant experiences of working with government agencies and financial institutions. Working experience with bilateral and multi-lateral development agencies will be given priority. Experience in planning, monitoring and evaluation of community-based projects will be an added advantage. S/he must be computer literate. Ability to speak and write in Nepal and English is required.

Terms of Reference (ToR)

Policy and Institutional Strengthening Expert

The Policy and Institutional Strengthening Expert (PISE) will work directly under the supervision and guidance of the NPM with the main responsibilities to coordinate with AEPC and SASEC to implement energy related activities in a harmonized way. Acting as the GESI Focal Point of the project, the PISE will also oversee the implementation of GESI related activities of the project to meet the agreed GESI targets. Specifically, PISE will:

- Support to prepare Annual and Quarterly Plans and budgets in coordination with respective AEPC components and SASEC
- Support to implement TA activities in close coordination with AEPC and SASEC.
- Support AEPC in planning and policy formulation including for promotion of mini hydro and solar installations
- Support in public sector, private sector and financial institution engagement in the SASEC project
- Support in effective implementation of activities as defined in GESI framework for projects
- Actively participate and provide GESI specific technical support to the AEPC.
- Design and deliver training and build capacity of staff on various perspective of GESI so that project personnel and relevant stakeholders take responsibility for implementing gender sensitive activities and interventions
- Provide inputs in preparing GESI policy and in revision of subsidy policy and delivery mechanism.
- Provide inputs to update project documentations from GESI perspective.
- In coordination with LE, work to identify business potential and cases that empower DAGs and women and feeding these into the development of innovative business cases/models in required activities.
- Maintain disaggregated data and devise the monitoring forms and formats from GESI perspective.
- Conduct regular monitoring of GESI activities and produce periodic reports.

Likewise, the PISE will also carry out the following tasks:

- Support AEPC to coordinate plans and activities of stakeholders related to Clean Cooking and Lighting Solutions for All, Scaling-up Renewable Energy, Sustainable Energy for All, etc.
- Support Municipalities and Rural Municipalities to prepare Municipal Energy Plans in close collaboration with NEA incorporating both grid and off grid solutions for effective and efficient use of resources.
- Support in institutionalization of mini hydro and solar and solar wind hybrid installations supported by AEPC/SASEC through formation and capacity enhancement of SPV.
- Support to identify gaps in dissemination of mini hydro and solar and solar wind hybrid systems and prepare capacity development package.
- Participate in different forums organized by various agencies related to RE, PEU, poverty alleviation, livelihoods and GESI.
- Assist in networking amongst the key stakeholders of RE.
- Help AEPC to institutionalize RE based best practices and models established.
- Documentation and sharing of the lessons learnt from the different innovative technologies and implementation process of promotion of RE with other stakeholder in national as well as regional level.

Qualifications: The PISE must possess a Master's degree in Renewable Energy, Engineering or related field with preferably 10 years' of work experience in RE sector. The PISE shall have knowledge of specific gender issues in renewable energy and sensitivity to local culture and traditions in Nepal and also possess demonstrated capacity for organization and management with excellent reporting and coordination skills. Preference will be given to previous experiences in policy and planning for promotion of. Experience on mini hydro, solar mini grid system and PEU will be an advantage.

Terms of Reference (ToR)
Mini Hydro Expert

The Mini Hydro Expert (MHE) will be responsible for providing technical support and leadership in areas pertaining to mini hydro development and operation. The MHE will closely work under the overall guidance of the NPM. Specifically, the MHE will carry out the following activities related to off-grid mini hydro:

- Mobilize private sector, BFIs as well as local authorities to participate in the off-grid mini hydro development.
- Conduct study on identification of technical challenges and opportunities in design, manufacturing, installation and after-sales service in mini hydro.
- Support in improving capacities project identification, feasibility study, detailed design, manufacturing, construction, installation, O&M through development of manuals, capacity building activities, trainings, guidelines etc.
- Review and update the detailed feasibility reports of the subprojects in close coordination with the NPM and relevant experts.
- Support in obtaining permits and clearances.
- Support in the formation of Special Purpose Vehicles (SPV) and in the preparation of their business plans.
- Assist in implementation of approved projects by providing technical and procurement assistance, installation assistance, and appropriate capacity building activities.
- Support to develop tariff collection modality and analysis for threshold level of tariff rate for the project to sustain.
- Support to develop operation and management (O&M) modalities.
- Monitor operations to assess progress, identify problems, document lessons learned.
- Identify and seek corrective measures for emerging problems.
- Carry out other activities to ensure the success of mini hydro subprojects.
- Support local governments to incorporate mini hydro into Municipal Energy Plans (MEP)
- Support revision of subsidy policy to attract more private sector investment.
- Assist in attracting investment in mini hydro.
- Develop specifications, in close collaboration with Nepal Electricity Authority (NEA), for mini hydro that meet national codes and standards for connection to the national grid.

Qualifications: The MHE must possess a Master's degree in Renewable Energy, Civil, Mechanical, Electrical Engineering or a related field. The MHE should preferably have 10 years of practical experience working in mini hydro development. Preference will be given to work experience in mini hydro, other RE and PEU related development works in Nepal.

Terms of Reference (ToR)

Solar Power Expert

The Solar Power Expert (SPE) will be responsible for providing technical support in areas pertaining to electrical components of mini hydro, solar mini grid and solar-wind hybrid mini grid system development and operation. The SPE will closely work under the overall guidance of the NPM. Specifically, the SPE will:

- Mobilize communities, private sector, BFIs as well as local authorities to participate in solar mini grid, hybrid mini grid, solar pumping, and institutional PV development
- Provide support in the designing and development of distribution networks in subprojects
- Provide support to make appropriate changes in policy mechanisms to facilitate development and operation of solar mini grid, solar pumping, and institutional solar PV.
- Develop specifications, in close collaboration with NEA and AEPC, for mini hydro, solar mini grid and solar/wind hybrid systems that meet national codes and standards for grid interconnection to the national grid.
- Support in incorporating solar mini grid and hybrid systems into MEP.
- Conduct study to identify the technical challenges and opportunities in design, installation and after-sales service in mini grid systems.
- Support in improving capacities of project identification, feasibility study, detailed design, installation, operation and management through development of manuals capacity building activities, trainings, guidelines, etc.
- Prepare detailed feasibility reports of the subprojects in close coordination with the NPM, AEPC and other relevant experts.
- Support in preparation of Detailed Project Report.
- Support the formation of SPV, where required, and in the preparation of their business plan.
- Assist in implementation of approved subprojects by providing technical, procurement and installation assistance, and appropriate capacity building activities.
- Support to develop O&M modalities.
- Monitor operations to assess progress, identify problems, document lessons learned.
- Identify and seek corrective measures for emerging problems.
- Carry out other activities to ensure the success of mini hydro, solar mini grid and hybrid systems.

Qualifications: The SPE must possess a Master's degree in Electrical Engineering, Renewable Energy, Energy or related Engineering field. The REE should preferably have 10 years of practical experience working in solar power development. Preference will be given to working experience in mini grid related development works in Nepal.

Terms of Reference (ToR)
Livelihood Expert

The Livelihood Expert (LE) shall be responsible for sustainable promotion of PEU of electricity. She/he will be responsible for overall community mobilization and institution development of RE projects. The LE will closely work under the overall guidance of the NPM. Specifically, the LE will:

- Work closely with the project team to promote PEU in RE projects.
- Work in close coordination with the other experts to develop instruments to incentivize financial institutions to promote financial products to entrepreneurs and end users.
- Coordinate with potential partners and business development services providers for promotion of PEU in aspects such as awareness creation, capacity building, financing and technology.
- Support the project developers to establish appropriate institution cooperative/company to install and manage energy projects
- Facilitate to develop rules, regulation guidelines and manuals for sustainable operation of the system and orient and support to implement the operational and management guidelines of power projects
- Prepare guidelines for assessing existing businesses in the electrification area having potential for switching to electricity.
- Support the RE project developers in setting up an Enterprise Promotion Unit within its management set up and preparing business plan for promoting PEU.
- Support existing entrepreneurs for switching to electrical energy.
- Conduct exploration study for identifying potential and feasible enterprises in the subproject area.
- Conduct business promotion campaigns.
- Provide capacity building support to entrepreneurs to establish new enterprises.
- Facilitate to enhance the capacity of project developers, management committees, and entrepreneurs for sustainable operation of the power projects
- Provide capacity building support to existing entrepreneurs for smooth operation of the business, business expansion and productivity improvement.
- Document and showcase successful enterprises for replication.
- Select electricity-based enterprises for possible technical as well as financial support; facilitate entrepreneurs to take advantage of subsidy provision.
- Ensure gender sensitization at every stage of business promotion process, from awareness creation to capacity building, business creation and operation.

Qualifications: The LE must possess a Master's degree in Business Administration, Development Studies, Social Science, Economics or a related field. The LE should have preferably 10 years of practical experience working in promotion of rural enterprises for livelihood enhancement and 5 years successful technical assistance in renewable energy sector particularly in productive energy uses and local community institution development and strengthening. The LE must have knowledge of all major sectorial issues and possess analytical skills in planning and management of development programs.

Terms of Reference (ToR)
Administration and Finance Officer

The Administration and Finance Officer (AFO) will work directly under the supervision and overall guidance of the NPM. The AFO will be responsible for ensuring smooth administrative, human resource and financial management of the project. S/he will also be responsible for smooth information flow and human resource development. S/he will directly supervise the project support staff. Specifically, the AFO will:

- Maintain accurate financial and personnel records of the programme as required by GoN, ADB and UNDP.
- Supervise administrative and financial records of the staffs as needed.
- Assist the NPM in all matters related to financial and administrative functions of the project.
- Manage all correspondence related to administrative, financial and security matters.
- Maintain the inventory and keep records of all office equipment, furniture, fixing, and vehicle.
- Undertake physical verification of all assets and update records.
- Work as a resource person and trainer on administrative and financial management to train project staffs and other stakeholders.
- Undertake all tasks related to the audit and financial management.
- Carry out additional support as requested by the TL and as required to make this programme a success.
- Carry out all the administrative coordination with all project activities
- Provide support to NPM for financial planning and budget revisions.
- Monitor and update financial delivery and timely inform NPM and the team to take corrective actions.
- Assist the NPM in preparing progress reports, situation reports and impact studies on periodic manner.
- In a timely manner, prepare annual work plan, budget revision, quarter work plan and financial reports to be submitted to UNDP.
- Provide support to NPM in the implementation of human resource development initiatives, organize HRD related training, workshops, seminars and orientation to community people and staff of all key stakeholders on disaster risk reduction/management.

Qualifications: The AFO shall have a Master's degree in Business Administration or any other relevant field with preferably 7 years' work experience. The incumbent shall have experience in office management, procurement, organization of training programs and provision of secretarial and communication services. Proven abilities in English writing and computer skills are required. The candidate must have previous experience of working in UNDP financed or implemented projects.

Terms of Reference (ToR)
Knowledge Management Officer

The Knowledge Management Officer (KMO) will work directly under the supervision and overall guidance of the NPM and closely liaise with respective technical experts. S/he will be responsible for quality control and regular monitoring of the project interventions. The KMO will also work closely with AEPC in generating knowledge products of the project and sharing information to the larger stakeholders of the project. Specifically, the KMO will:

- Support database management of information collected by the project
- Carry out necessary analysis and prepare summary reports
- Support the team, AEPC and ADB to establish appropriate monitoring systems in place.
- Prepare monitoring schedules of the project activities as required by AEPC, ADB and UNDP.
- Assists Technical Experts in the preparation of monitoring and evaluation reports as needed.
- Ensure regular back-up of MIS database and electronic documents stored in computers.
- Assist in preparing data and analysis in proper GIS presentations in the programme reports and studies.
- Assist in preparing knowledge products of the project including audio, visual, print, web-based materials.
- Carry out additional support as requested by the TA and NPM and as required to make this programme a success.

Qualifications: The KMO shall have a Bachelor's degree in Arts, Commerce, Economics or related fields with preferably 7 years' work experience in monitoring, database management, analysis and reporting in the RE sector. S/he shall have experience in producing and disseminating various knowledge products. Proven abilities in report writing in both English and Nepali are required. Preference will be given to candidates with experience on GIS software.

Terms of Reference (ToR)
Environment and Social Safeguard Expert

The Environment and Social Safeguard Expert (ESSE) will ensure the implementation of environment/social safeguard activities in line with available applicable frameworks. The ESSE shall be responsible for management, oversight, and monitoring of implementation of ESS compliance of the project during preparation and implementation phase.

The Expert will support the Project Implementation Unit to conduct, including but not limited, to the following tasks:

- Implement Environmental Assessment and Review Framework (EARF), Indigenous People's Planning Framework (IPPF) and Resettlement Framework (RF) of the project
- Prepare and implement plan for environmental training/orientations, and train/orient the project staff and relevant stakeholders on the application of Frameworks
- Prepare and/or review EMP/IEE/due diligence reports prepared for the projects
- Review the DFS/DED as well as bid documents of each project and verify that environmental mitigations, recommended in EMP/IEE/due diligence reports, are incorporated during project planning and implementation
- Provide guidance to project developers in preparing the IEE for each project
- Guide, advise and ensure that the contractors/developers consult affected people/stakeholders in accordance with EARF, RF and IPPF during implementation
- Ensure that the environmental safeguard compliance monitoring is conducted and reported as per National regulations and ADB's requirement
- Carry out GESI activity monitoring and ensure compliance to GESI targets and Action Plan
- Prepare the Grievance Redress Mechanism (GRM) and its handling procedures, channels based up on EARF and prepare GRM database to ensure that all registered complaints are addressed properly
- Prepare Semi-annual Environmental Compliance Monitoring Report and Semi-annual Social Monitoring Report
- Any other tasks assigned by NPM to support the project with respect to environmental and social issues.

Qualifications: The ESSE shall at least have a Master's degree in Environmental Science, Environment Management, Environmental/Civil Engineering or related fields with preferably 7 years' work experience related to environment and social safeguard. S/he shall have 3 years' experience related to environment and social safeguard in RE sector and proven experience in ensuring environmental and social safeguard compliance in mini/small hydro and solar mini grid projects in Nepal. S/he shall have working experience in projects supported by different development partners preferably with UNDP, ADB and/or World Bank. Proven abilities in report writing in both English and Nepali are required. This position is requires ability to visit remote parts of the mini hydro and solar mini grid sites in remote areas of the country on a regular basis.